



PE CIVIL-STRUCTURAL EXAM 06

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Problems

[Click here for Solutions](#)

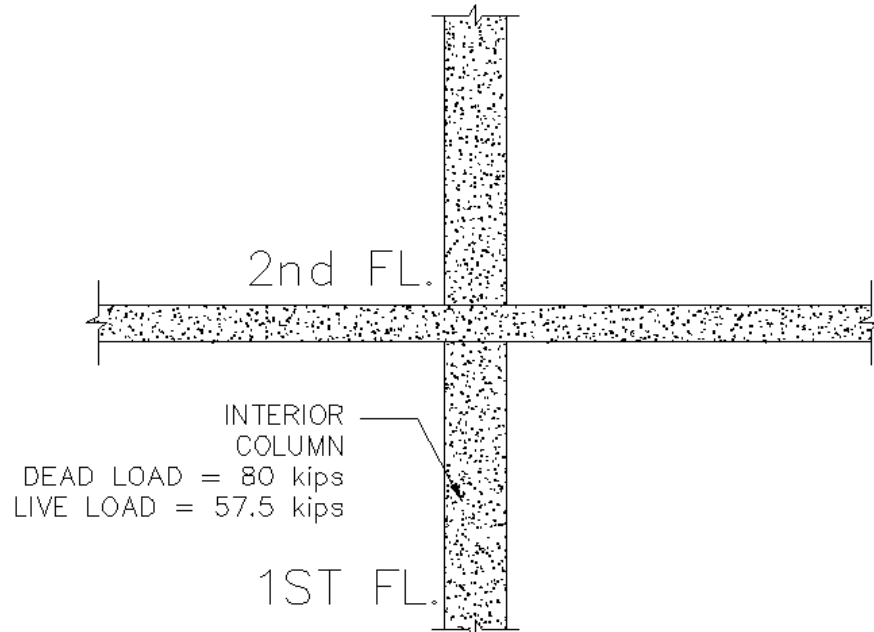
1. A residential building is undergoing underpinning and retrofitting of the foundation. The total dead load on the East wall is calculated as 2020 kN. The wall must be jacked up but the ratings on the hydraulic ram are only available in US (short) tons only. Select the correct rating of the jack that can be used for this job to satisfy the minimum requirement.

 - (A) 100 tons
 - (B) 200 tons
 - (C) 225 tons
 - (D) 250 tons

2. A timber beam at the roof of a building supports a parapet wall along its length. The wall is 12 ft long and 4 ft tall. The wall is constructed with hard brick masonry and is 9 inches thick. What is the factored dead load to be used in combination with live loads, due to the masonry's weight acting on the beam?

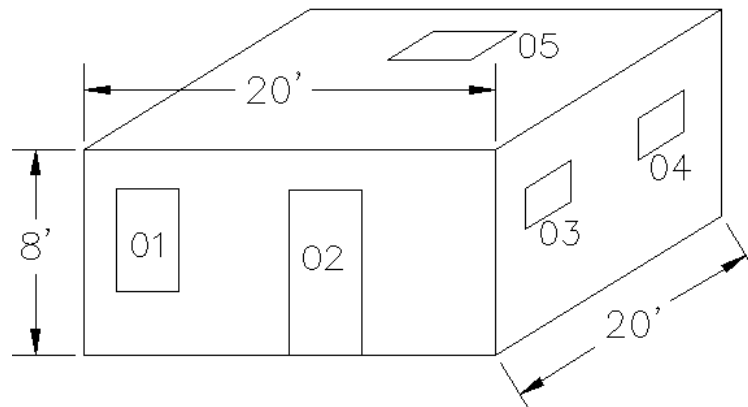
 - (A) 390 lb/ft
 - (B) 468 lb/ft
 - (C) 561 lb/ft
 - (D) 673 lb/ft

3. A three-story retail store has a heavily loaded interior column on the first floor. The design is to be optimized, and the engineers are looking to reduce the live load demands on the column. The tributary area of the column is 500 ft^2 . Calculate the reduced live load demand acting on the column.



- (A) 20 kips
(B) 25 kips
(C) 34 kips
(D) 40 kips
4. ASCE 7-16 Table 4.3-1 provides minimum distributed live loads and minimum concentrated live loads to be used for design for structures. Which of the following is not true about concentrated loads?
- (A) Concentrated live loads are assumed to act uniformly over an area of 0.58 m^2
(B) ASCE 7-16 specified concentrated live loads are summed with minimum uniformly distributed live loads to create the most severe effect in the structural member
(C) Live load reduction factors cannot be applied to concentrated loads.
(D) Concentrated loads are usually critical over uniformly distributed loads during slab design.

5. A small shed shown below is to be constructed near a coastal area. The building has openings in two walls and roof as shown below. The other two walls are solid and has no openings



The sizes of openings are as follows

O1 = 2ftX3ft O2 = 2ftX6ft

O3 = 2ftX2ft O4 = 2ftX2ft

O5 = 4ftX4ft

The front wall with openings O1 and O2 receives positive wind pressure.

Based on the information above, the shed can be categorized as?

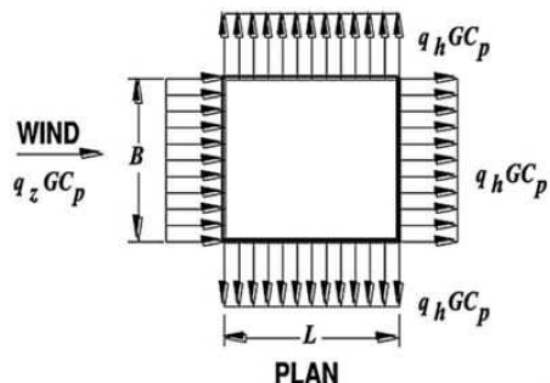
- (A) Enclosed
 - (B) Open
 - (C) Partially enclosed
 - (D) Partially open
6. Using the directional procedure, compute the design wind pressure in the windward walls of a rigid enclosed 1 story building located in Exposure C terrain. The basic wind speed of the area is 100 mph. Neglect any ground elevation and topographic effects.

L = 30ft

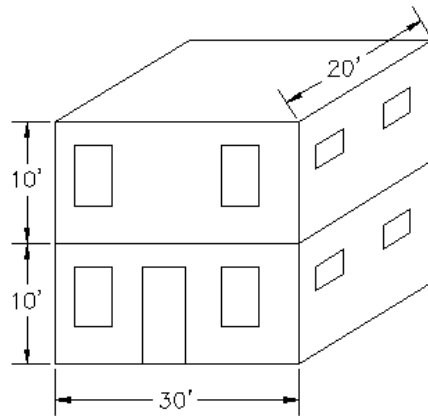
B = 30ft

Height of wall $z = h = 10$ ft

- (A) 9.25 psf, 15.91 psf
- (B) 15.91 psf, -9.25 psf
- (C) 9.25psf, -15.91 psf
- (D) -9.25psf, -15.91 psf

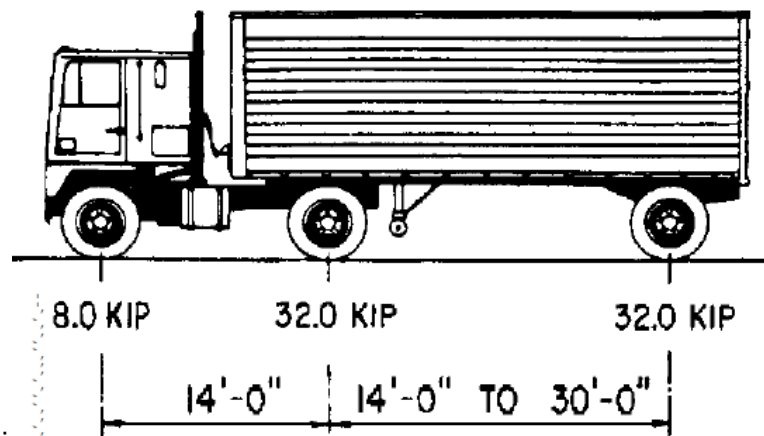


7. A 2 story ordinary plain masonry wall building is identified as seismic design category B. The mapped short period spectral acceleration $S_s = 0.5g$. The site is rocky and the total seismic weight of the building is 600 kips. Calculate the seismic base shear using simplified lateral force analysis procedure per ASCE 7-16.



- (A) 100 kips
(B) 150 kips
(C) 250 kips
(D) 300 kips

8. The figure below shows a standard design Truck from AASHTO LRFD bridge design specifications. What is the pressure exerted by tires on each side of the rear axle?



- (A) 40 psi
(B) 60 psi
(C) 80 psi
(D) 160 psi

9. Calculate the design snow loads acting on the roof of a building located in Anchorage Alaska. The building is classified as Risk Category III, The roof is unheated and partially exposed. The site is in an urban area with numerous closely spaced obstructions.

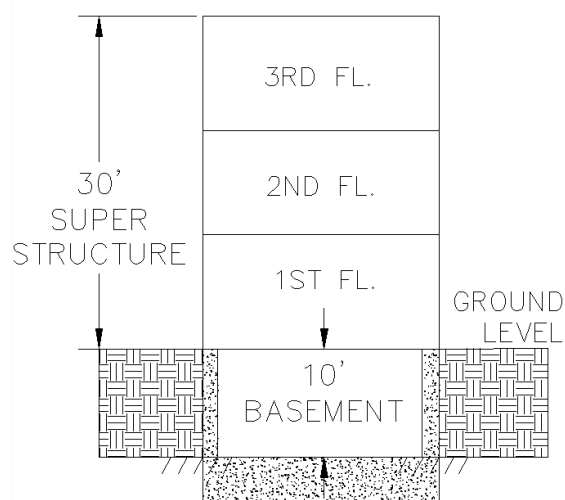
(A) 42.6 psf
(B) 46.2 psf
(C) 50 psf
(D) 55psf

10. Which of the following is incorrect about ice sensitive structures and their design per ASCE 7-16?

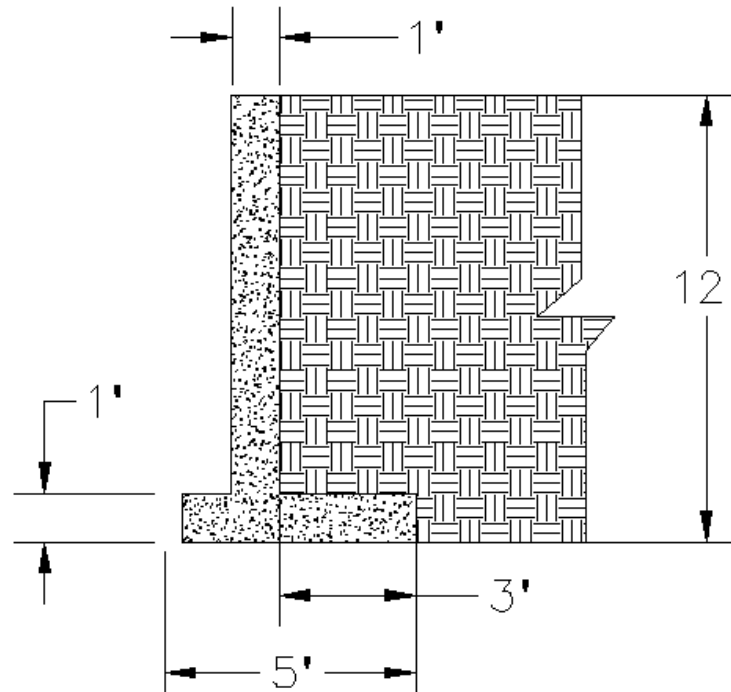
(A) In areas with freezing rain and snow, lattice structures, guyed masts, overhead lines, light suspension and cable-stayed bridges, aerial cable systems are sensitive to ice loads,
(B) While analyzing a structure for ice loads, the load due to weight of ice and the load due to wind on ice formed areas are considered separately to avoid conservatism in design,
(C) For temperature-sensitive structures, the load shall include the effect of temperature change from everyday conditions to the design temperature for ice and wind on ice,
(D) Dynamic load effects due to atmospheric icing are not covered in ASCE 7-16.

11. A three story building shown below has a 10 ft deep basement parking level. 12 inch thick reinforced concrete foundation walls extend from the mat to the 1st floor slab level. The soil retained by the foundation wall is well graded gravel-sand mixes. Calculate the total lateral load exerted by the soil on the East foundation wall. The total length of the East wall is 40 feet. Water table can be assumed below the foundation mat.

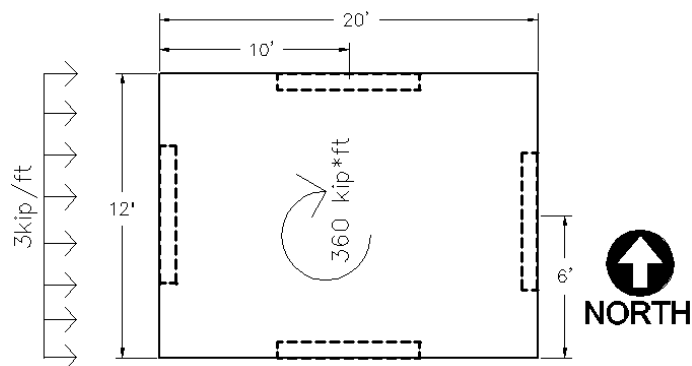
(A) 60 kips
(B) 90 kips
(C) 120 kips
(D) 150 kips



12. A reinforced concrete retaining wall shown below has backfill on one side of the wall. Calculate the factor of safety against overturning.
- Density of soil, $\gamma = 115$ pcf
- Density of concrete, $\gamma_c = 150$ pcf
- Active earth pressure coefficient = 0.33

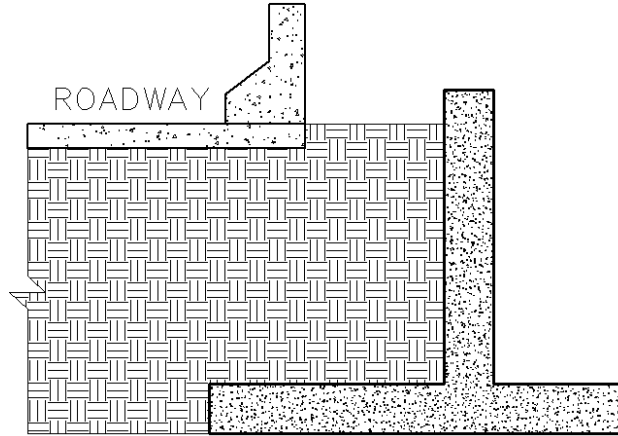


- (A) 1.2
- (B) 1.4
- (C) 1.6
- (D) 1.8
13. The rigid diaphragm shown below has a lateral load and a torsional moment acting at the center. The center of rigidity is at the geometric center of the diaphragm. The in-plane stiffness of each wall is equal. What is the maximum shear load on an individual wall?



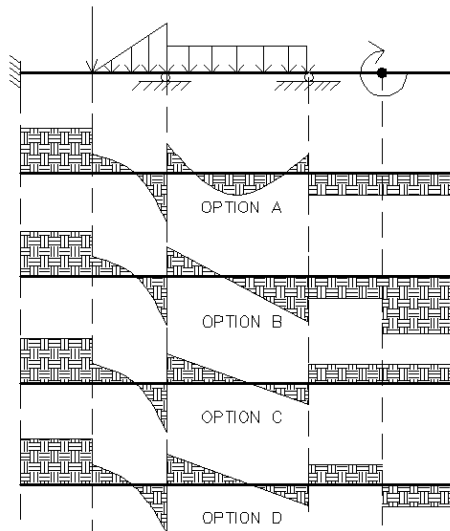
- (A) 18 kips
- (B) 24 kips
- (C) 26 kips
- (D) 30 kips

14. Partial section of an elevated roadway made on an embankment is shown below. The moment due to the back fill's unfactored active earth pressure on the wall is 10 kip*ft/ft. The moment due to live load surcharge from a vehicle travelling parallel to the wall is 5.5 kip*ft/ft. Moment due to wind load on the exposed stem of the wall above the backfill is 1.5 kip*ft/ft



What is the maximum factored moment acting on the wall per Strength V combination?

- (A) 20 kip*ft/ft
 - (B) 22 kip*ft/ft
 - (C) 24 kip*ft/ft
 - (D) 26 kip*ft/ft
15. For the given three span beam with a hinge at the center of the third span, what is the correct shear force diagram?

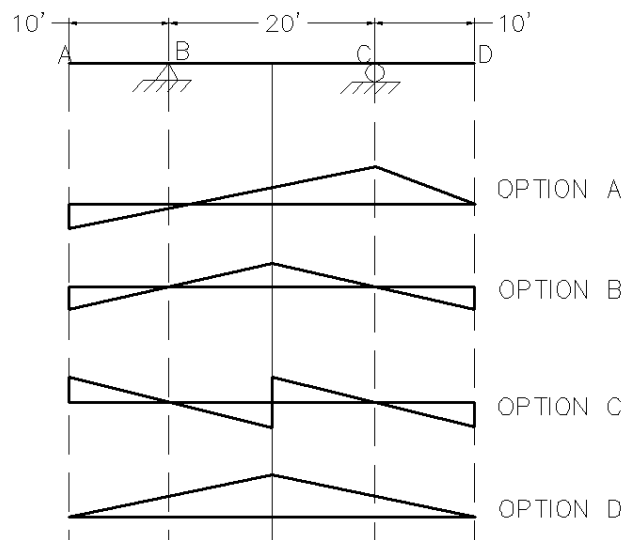


- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

16. Which of the following is not correct about shear force and bending moment diagrams?

- (A) Shear force between two unloaded points in a beam is constant,
- (B) The slope of the shear force diagram at any point gives the intensity of moment at that point,
- (C) The area between the shear force diagram between two points gives the change in moment between those points,
- (D) Maximum bending moment occurs at the point where shear force is zero.

17. For the beam shown below, select the correct influence line diagram for maximum moment at midspan, when a concentrated load passes from point A to point D

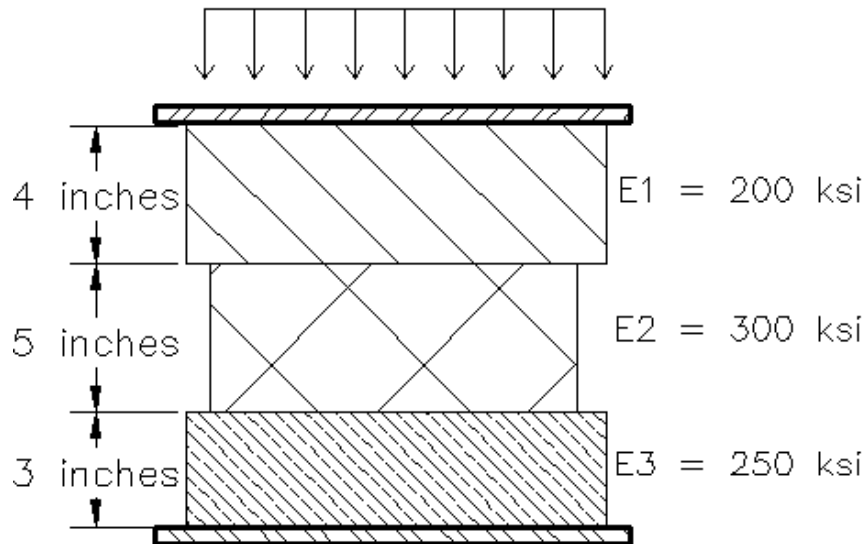


- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

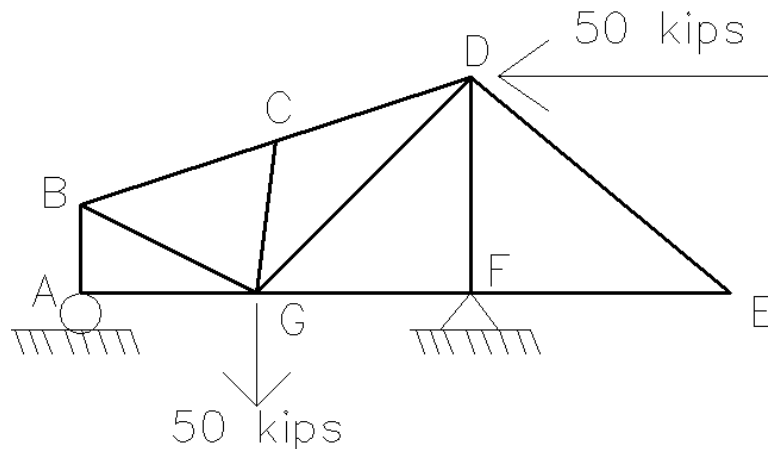
18. A cable is suspended between two points that are 50 ft high. The only load that is acting in the cable is its own weight. What is the shape of the cable?

- (A) Parabolic Curve
- (B) Linear
- (C) Catenary curve
- (D) Circular arc

19. A bearing system is developed using three materials as shown below. The cross sectional areas of the top and bottom layers are 20 in^2 and the middle layer is 16 in^2 . What is the total axial stiffness of the bearing system?



- (A) 378.5 kip/in
 (B) 588.5 kip/in
 (C) 1000.75 kip/in
 (D) 3626 kip/in
20. Find the number of zero force members in the truss shown below. Vertical load is acting on point G and horizontal load is acting on point D.



- (A) 2
 (B) 3
 (C) 4
 (D) 5

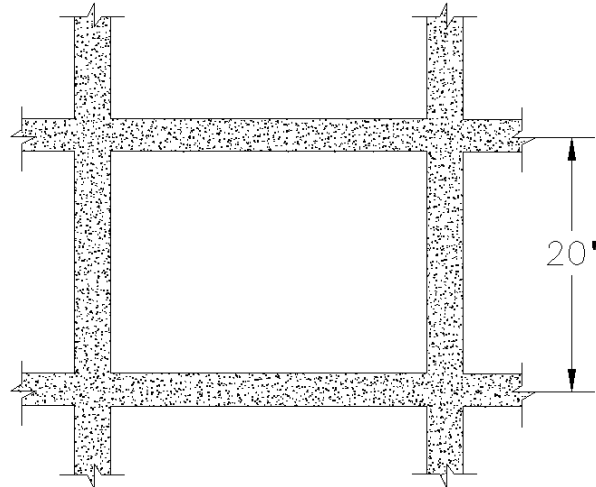
21. The frame below shows partial elevation of beams and columns in an intermediate moment resisting, reinforced concrete system. The center to center vertical distance between the beams i.e. center to center height of columns is 20ft. The depth of both top and bottom beams is 24 inches. The nominal flexural strength at the top and bottom of the columns is 450 kip*ft and 600 kip*ft. The unfactored shears acting on the columns are as follows:

Dead load shear = 5 kips

Live load shear = 10 kips

Earthquake load shear = 20 kips

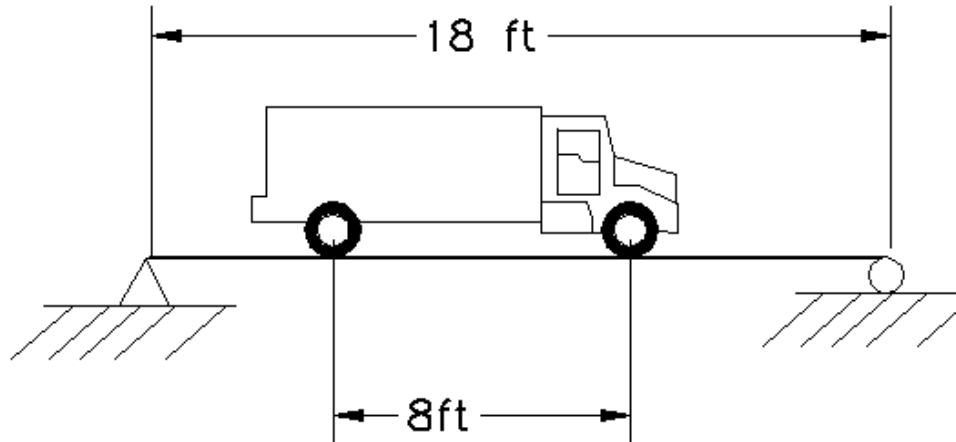
What is the factored design shear to be considered for the column?



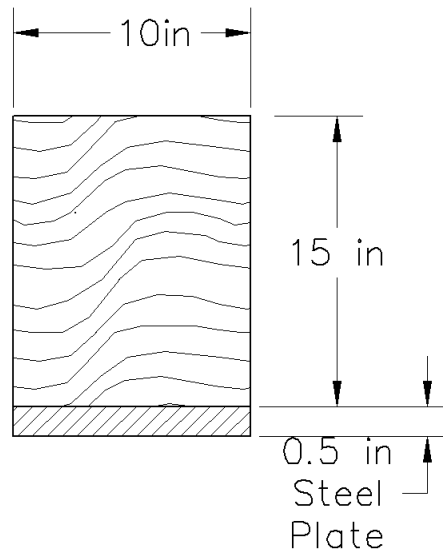
- (A) 45 kips
(B) 60 kips
(C) 76 kips
(D) 89 kips
22. The equation of bending moment in at a certain span of a beam system is $M = x^3 - 5x^2 - 3x$, where x is in ft and M is in kip*ft. What is the value of the shear force in that span when $x = 3$ ft?

- (A) -6 kips
(B) 6 kips
(C) 27 kips
(D) -27 kips

23. A truck travels along a simply supported beam. The distance between the front and rear axle is 8 ft. The span of the beam is 18 ft. If the front axle weighs 3 kips and the rear axle weighs 8 kips, what is the maximum possible shear in the beam without considering dynamic load effects?



- (A) 3 kips
(B) 8 kips
(C) 9.67 kips
(D) 11 kips
24. A composite beam constructed using wood, reinforced with 0.5 inch thick steel plate at the bottom as shown below is subjected to a bending moment of 150 kip*ft. What is the flexural stress acting on the steel? Modulus of elasticity of wood = 2500 ksi and modulus of elasticity of steel = 30000 ksi.

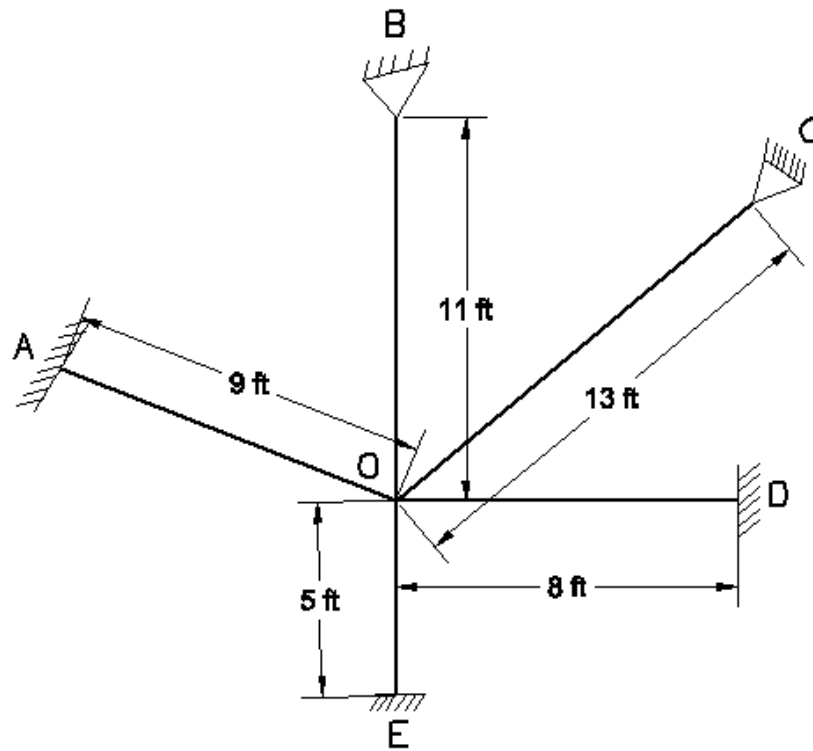


- (A) 23.2 ksi
(B) 25.3 ksi
(C) 30.6 ksi
(D) 33.8 ksi

25. Which of the following is not true about plastic section modulus ?

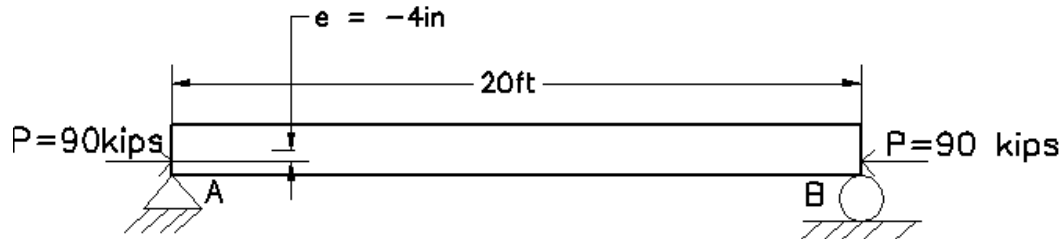
- (A) The shape factor, i.e. the ratio of plastic to elastic section modulus of I sections is lower than rectangular sections,
- (B) Structures designed using plastic section modulus is generally lighter than structures designed using elastic section modulus,
- (C) When calculating plastic section modulus, the neutral axis divides the areas of compression block and tension block equally,
- (D) Plastic section modulus is always independent of the material property.

26. For the framing configuration below, a correction moment of 50 kip*ft is applied at joint O. What proportion of correction moment is distributed to member OB? The moment of inertia and modulus of elasticity of all the sections are uniform.



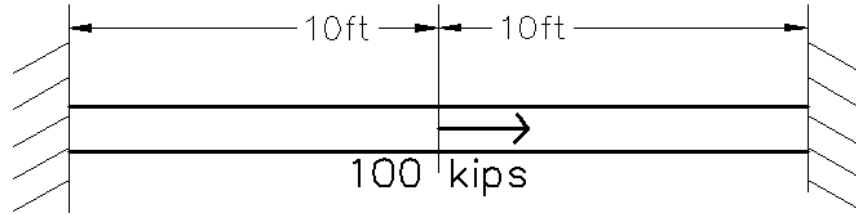
- (A) 5.23 kip*ft
- (B) 6.07 kip*ft
- (C) 9.18 kip*ft
- (D) 15.34 kip*ft

27. A beam is subjected to a prestressing force of 90 kips applied 4 inches below the centroid of the section. A uniformly distributed load of 3 kip/ft acts along the entire span of the beam. The area of the beam section is 240 in^2 and the section modulus is 800 in^3 . What is the maximum normal stress at the top fibre of the beam due to combined Uniform load and prestress ?

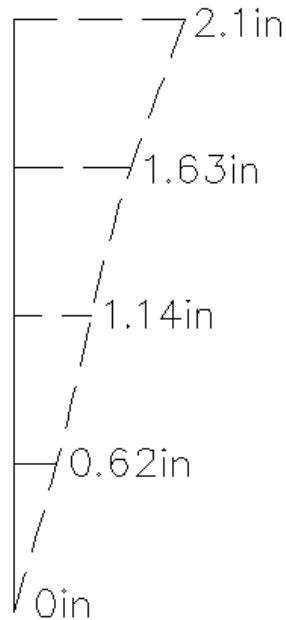


- (A) 1425 psi compressive
 (B) 1425 psi tensile
 (C) 2175 psi tensile
 (D) 2175 psi compressive
28. A solid circular shaft is subjected to a torsional moment of 10 ft-kips. The shaft is also subjected to a bending moment of 15 kip*ft. What is the equivalent twisting moment that when acting alone, produces the same maximum shearing stress as the combined effect of both moments acting currently?
- (A) 15 kip*ft
 (B) 18 kip*ft
 (C) 25 kip*ft
 (D) 28 kip*ft

29. A fixed-fixed beam with area 11 in^2 is subjected to an axial load at the center of the span. The beam was installed at 32°F with no room for expansion and the ambient temperature at service is 102°F . What is the total combined compressive stress acting in the beam section? Modulus of Elasticity = 29000 ksi Coefficient of thermal expansion = $0.000007 / ^\circ\text{F}$.



- (A) 4.54 ksi
(B) 9.09 ksi
(C) 18.76 ksi
(D) 20.34 ksi
30. A four story steel special moment resisting frame building is identified as an essential infrastructure and a seismic design category C. The elastic deflections of the floors are as follows. What is the maximum design story drift per ASCE 7-16?



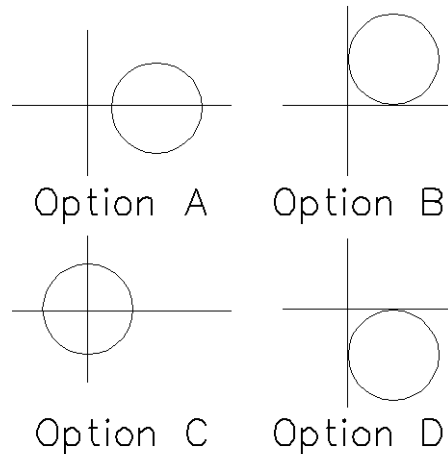
- (A) 2.1 in
(B) 2.27 in
(C) 2.5 in
(D) 2.78 in

31. The average story height of a three story steel frame building is 11ft. It is identified as Seismic Risk category III and seismic design category D. The structure has extreme torsional irregularity. What is the allowable story drift per ASCE 7-16?

- (A) 2 in
- (B) 2.64 in
- (C) 3 in
- (D) 3.2 in

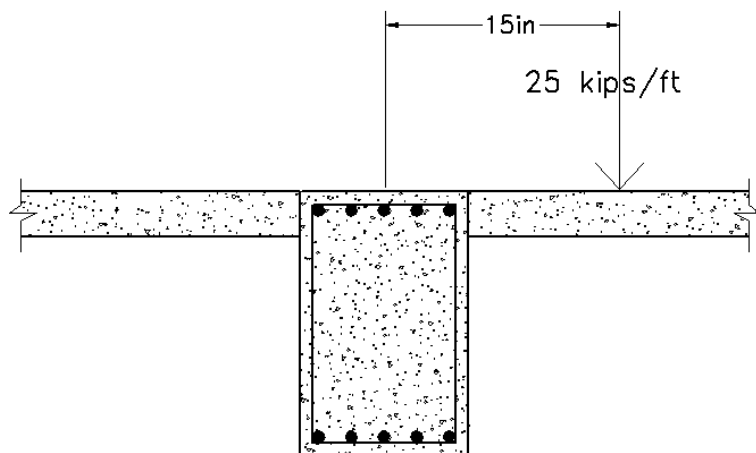
32. What is the shape of Mohr's circle for pure torsion?

- (A) Option A,
- (B) Option B,
- (C) Option C,
- (D) Option D

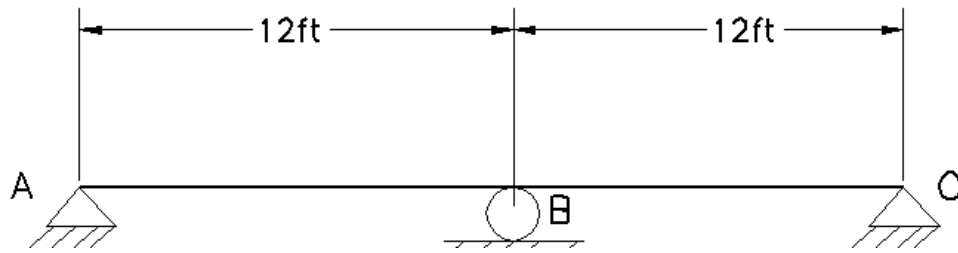


33. A reinforced concrete beam has a span of 20 ft. The beam is subjected to a live load of 25 kips/ft at a distance 15 in from the center of the beam as shown below. The beam is restrained against torsion only at one end. What is the torsional moment acting on the beam?

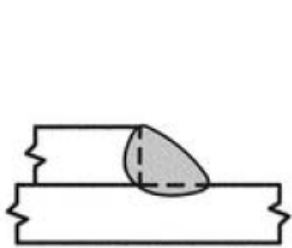
- (A) 125 kip*ft
- (B) 250 kip*ft
- (C) 312.5 kip*ft
- (D) 625 kip*ft



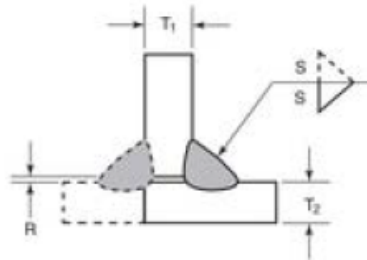
34. A continuous beam shown below has equal span lengths of 12 ft. What is the Moment developed at support B due to a settlement of 2 inches at support A? Modulus of elasticity = 29000 ksi, Moment of inertia = 1200 in^4



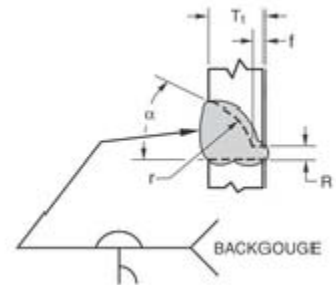
- (A) 380.8 kip*ft
 (B) 419.5 kip*ft
 (C) 450.7 kip*ft
 (D) 510.2 kip*ft
35. Which of the following welds require special inspections for joint preparation prior to welding?



Option I



Option II



Option III

- (A) Option I
 (B) Option II
 (C) Option III
 (D) Option I and II

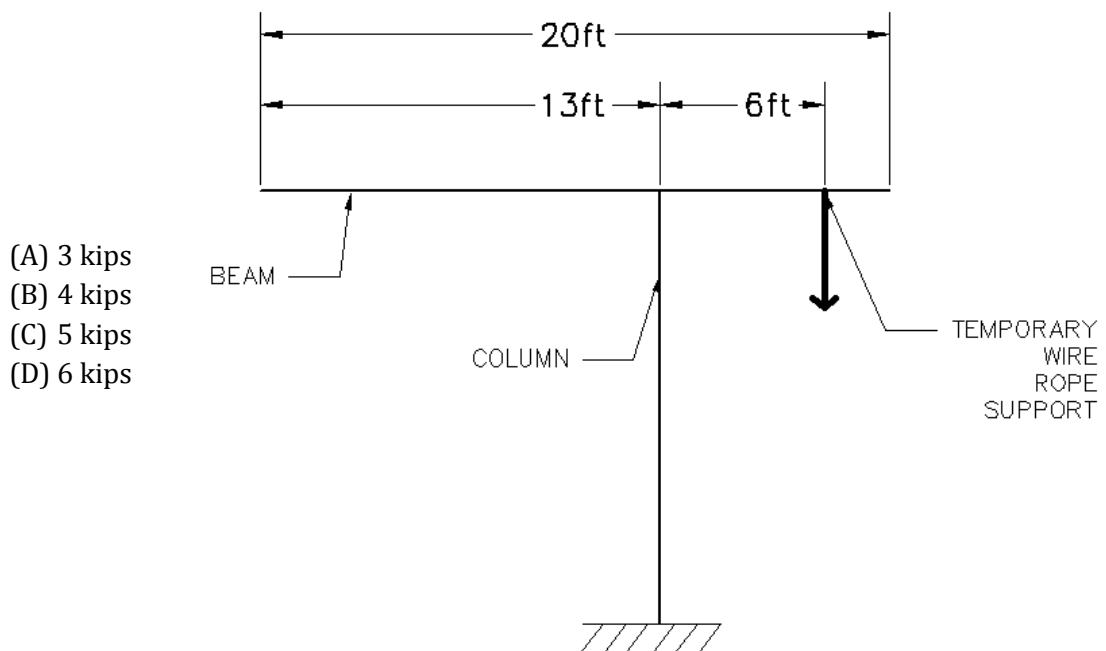
36. Which of the following wooden truss elements are subjected to the requirements of special inspection per IBC chapter 17

- (A) A wooden Pratt truss with overall height of 3 feet,
- (B) Site built King truss with a span 65 feet and metal plate connections,
- (C) Site built Queen truss with a span 50 feet and metal plate connections,
- (D) Prefabricated truss with span of 80 ft, the fabricator approved to work according to IBC clause 1704.2.5.1

37. Which of the following is typically not a part of the general contractor's submittal package for a commercial building project?

- (A) Vendor provided flooring product data sheet to be used in the building,
- (B) Concrete cylinder break reports provided by the testing lab,
- (C) Structural calculations for construction loads,
- (D) Mill test reports.

38. During a steel erection job, a 20 ft long beam is set on top of a column as shown below. The beam is temporarily supported by wire rope behind the column. If the weight of the beam is 200 lbs/ft, what is the maximum reaction on the column?



39. Two adjacent buildings are to be constructed without setbacks from the property line. One of the buildings is 5 story tall and has a maximum inelastic response displacement of 7 inches per ASCE 7-16. Another building is 6 story tall and has a maximum inelastic response displacement of 9 inches. What is the required separation between the two buildings to prevent seismic pounding?

- (A) 9 in
- (B) 12 in
- (C) 16 in
- (D) 20 in

40. A wire rope bundle is used in a powered platform mechanism and the maximum static tension on the bundle including the platform's self weight is 10 kips. Per OSHA regulations, what is the minimum required rated strength of the wire rope bundle

- (A) 10 kips
- (B) 40 kips
- (C) 60 kips
- (D) 100 kips

41. The figure below shows the subsurface profile of a site. The groundwater is at EL. 58 ft. What is the effective stress at elevation 45 ft?

GRAVEL, 120 pcf	EL. 75ft
SILT A, 130 pcf 	EL. 60ft
SILT B, 128 pcf	EL. 55ft
SAND, 125 pcf	EL. 40ft
ROCK, 140 pcf	EL. 20ft
	EL. 0ft

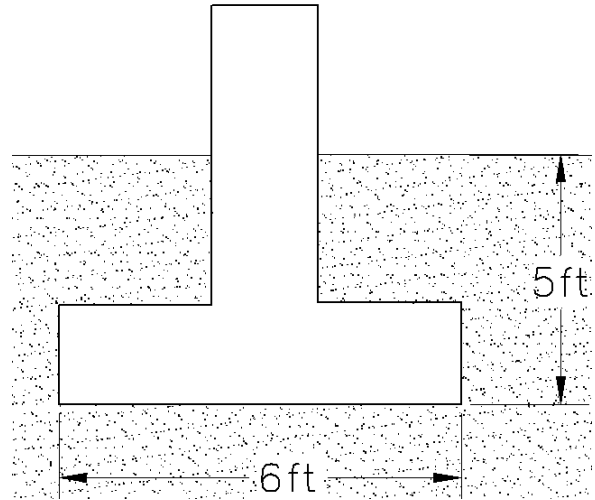
- (A) 2920 psf
- (B) 3110 psf
- (C) 3230 psf
- (D) 3400 psf

42. Which of the following is not true about liquefaction of soil?

- (A) Foundations built in submerged sandy soils are susceptible to liquefaction.
- (B) Clayey soils have higher cohesion and density and are immune to liquefaction.
- (C) During a strong earthquake, the pore water pressure increases and the soil loses its ability to support loads and liquefies.
- (D) Liquefied soil causes higher lateral earth pressures on retaining walls.

43. Calculate the net allowable bearing capacity for the square footing shown below. The factor of safety against bearing is 3. The groundwater table is observed at a depth of 25 feet. The density of soil is 115 pcf, the angle of internal friction is 20° and the cohesion = 350 psf.

- (A) 3.57 ksf
- (B) 4.33 ksf
- (C) 5.22 ksf.
- (D) 6.18 ksf



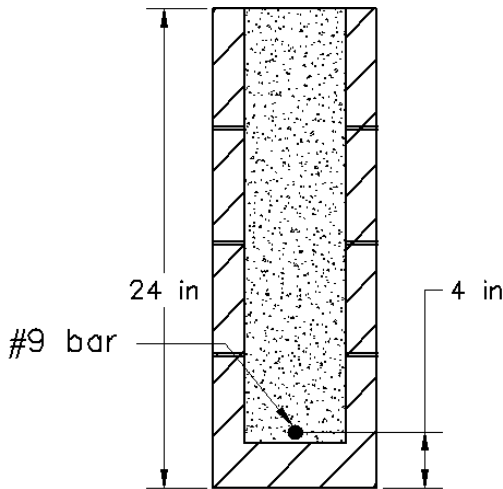
44. A concrete mix is designed for a roof slab that will be subjected to snow and ice accumulation and frequent rain. The nominal aggregate size is $3/4$ inch and the compressive strength of concrete is 4500 psi. What is the recommended water - cement ratio and total air content in the concrete?

- (A) Target air content 6 % and minimum w/c ratio 0.45
- (B) Target air content 5 % and minimum w/c ratio 0.55
- (C) Target air content 6 % and maximum w/c ratio 0.45
- (D) Target air content 5 % and maximum w/c ratio 0.55

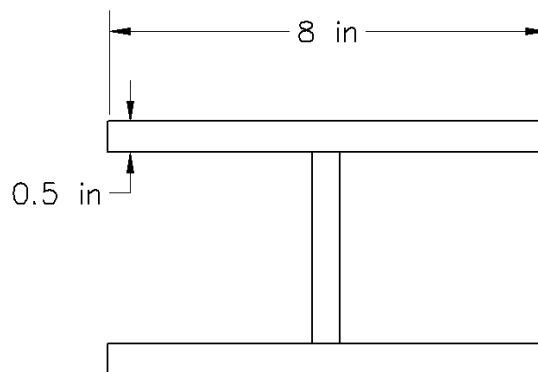
45. What is the modular ratio of a generic non prestressed rebar and a concrete of strength 8000 psi and density 155 lb/ft³?
- (A) 5.1
 - (B) 6.2
 - (C) 7.3
 - (D) 8.4
46. The gross area of a prestressed girder is 500 in². The profile of tendons are concentric with the gross area of the girder throughout the span and the jacking force is 350 kips. The concrete is normal weight concrete and the strength of concrete during initial prestressing is 5000 psi. What is the estimated loss in prestressing because of shortening of concrete?
- (A) 3.75 ksi
 - (B) 4.46 ksi
 - (C) 5.45 ksi
 - (D) 6.51 ksi
47. An I beam has flange thickness of 0.8 in and flange width of 7 in. Both flanges have two bolt holes, one on each side of the web. The diameter of the holes, adjusted per AISC 360-16 B 4.3b, is 0.875 in. What is the nominal moment capacity of the section?
- The elastic section modulus is 200 in³ and the plastic section modulus is 240 in³. The yield strength of the material is 50 ksi and the ultimate strength is 65 ksi.
- (A) 6500 kip*in
 - (B) 7820 kip*in
 - (C) 9750 kip*in
 - (D) 12000 kip*in

- 48.** Which of the following is true about the influence of specific gravity of wood on its mechanical properties?
- (A) Specific gravity of wood only affects thermal and moisture related properties of wood,
 - (B) Specific gravity of wood is directly proportional to the modulus of elasticity and inversely proportional to the modulus of rigidity,
 - (C) Specific gravity has negligible effects on mechanical properties except during dynamic loading conditions,
 - (D) Wood with higher specific gravity tends to swell and shrink more.
- 49.** A 4X10 Aspen No. 1 is to be used as a post. What is the ASD adjusted value of compression parallel to the grain for live load conditions? Assume 90°F temperature and wet service conditions. Take $C_i = 1$, $C_p = 1$.
- (A) 480 psi
 - (B) 540 psi
 - (C) 600 psi
 - (D) 700 psi
- 50.** What is the development length of a No. 6 bar used as tensile reinforcement in a masonry wall with specified strength 2000 psi ? The clear spacing between the bar splices is 3.5 inches and the minimum cover is 3 inches. All rebars used are Grade 60.
- (A) 12 in
 - (B) 36 in
 - (C) 43 in
 - (D) 48 in

51. The section below shows a fully grouted concrete masonry beam. The compressive strength of masonry is 2000 psi. The section is flexurally reinforced with one # 9 bar (Gr. 60). Determine the tensile stress on the rebar due to a moment of 25 kip*ft acting on the beam. Width of the section = 7.625 in

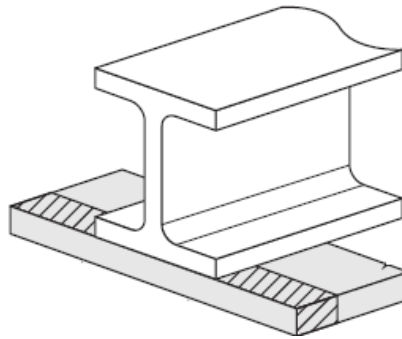


- (A) 11 ksi
(B) 13 ksi
(C) 17 ksi
(D) 21 ksi
52. A symmetric plate girder is constructed using A 572 Gr. 50 plates. The section of the girder is shown below. Based on the flange dimensions, which of the following is true about the compactness of the flange if the girder is subjected to axial loads and moments?



- (A) Compact
(B) Non Compact
(C) Slender
(D) Cannot be determined with given information

53. A simply supported Gr. 50 W30X90 beam is subjected to a uniformly distributed factored load of 10 kip/ft throughout the span of the beam. The span of the beam is 25 ft. What is the minimum length of the bearing plate 'lb' along the span of the beam?



- (A) 2.5 in
(B) 3.5 in
(C) 4.5 in
(D) 5.5 in
54. What is the maximum allowable concentrated load that can be applied to a Gr 50 ksi W10X30 section without the use of web stiffener plates. Assume load is applied sufficiently away from supports and length of concentrated load along the span = 1 in.

Section properties:

$$bf = 5.81 \text{ in}$$

$$tf = 0.51 \text{ in}$$

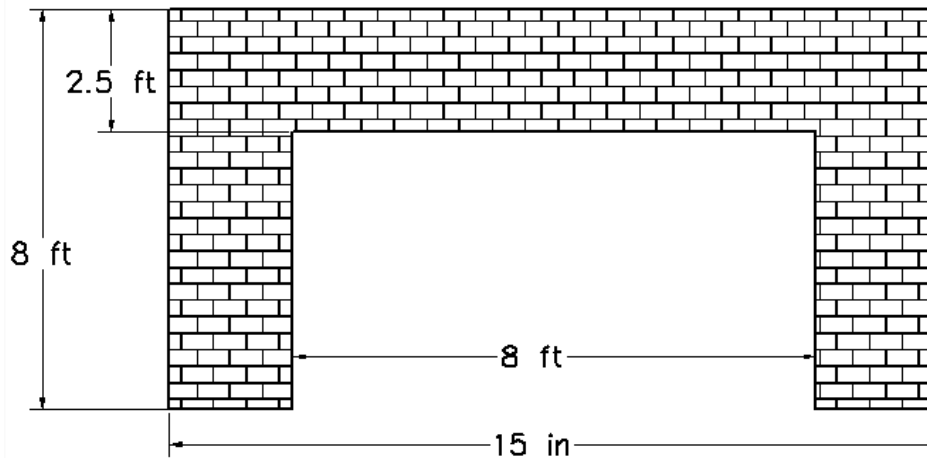
$$d = 10.5 \text{ in}$$

$$tw = 0.3 \text{ in}$$

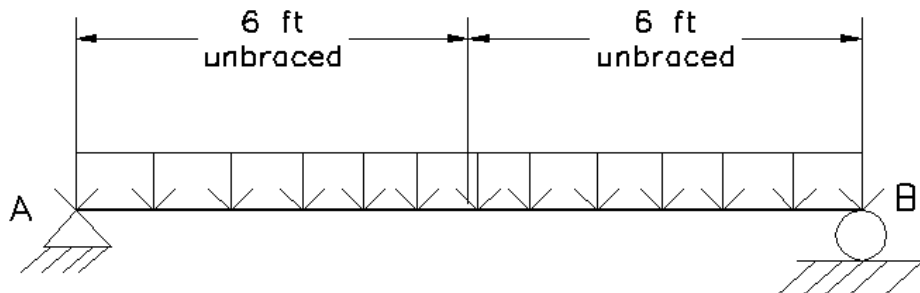
$$k = 0.81 \text{ in}$$

- (A) 48 kips
(B) 50 kips
(C) 70 kips
(D) 75 kips

55. What is the effective span of the masonry beam shown below? Beam is not built integrally with the piers.

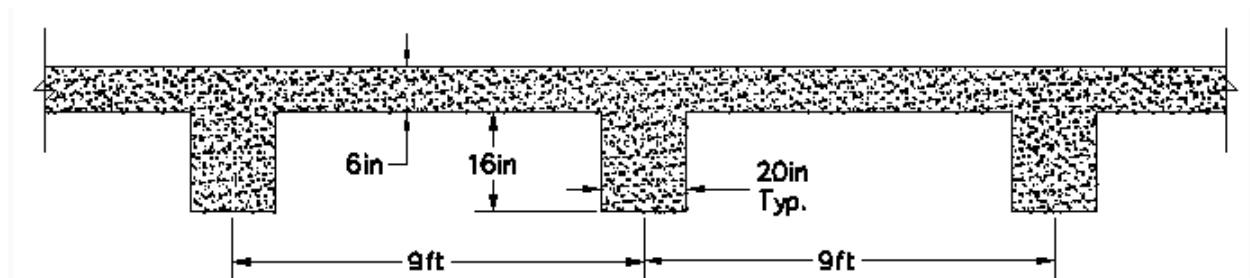


- (A) 8 ft
 (B) 10.5 ft
 (C) 11.5 ft
 (D) 15 ft
56. A simply supported 4X8 (nominal dimensions) southern pine section is subjected to a uniformly distributed load. The beam is braced laterally at midpoint. What is the beam slenderness ratio?



- (A) 7
 (B) 8
 (C) 9
 (D) 10

57. What is the effective T- beam flange width for the center beam shown below ? The span of the RCC beams are 30ft and the clear span is 27ft.



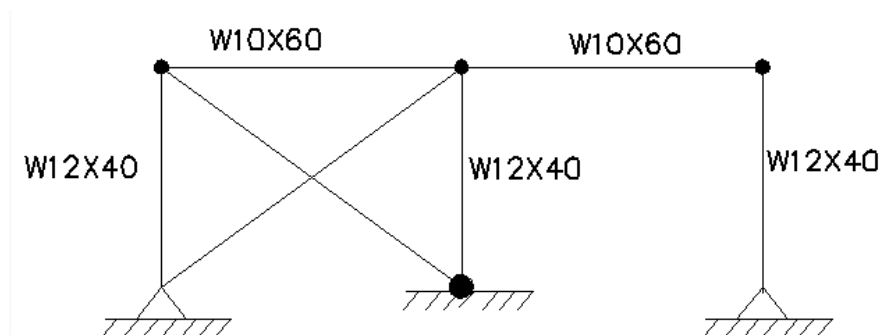
- (A) 48 in
- (B) 96 in
- (C) 101 in
- (D) 108 in

58. A 16 in X 18 in steel column base plate supporting a HSS 9X8X0.5 member is centered on a concrete pedestal. The dimensions of the pedestal are 2 ft X 2 ft. The strength of the concrete is 4 ksi and the steel plate is Gr 50. What is the maximum area of the concrete bearing frustum on which the bearing load is dispersed?

- (A) 288 in²
- (B) 528 in²
- (C) 576 in²
- (D) 1152 in²

59. The frame shown below belongs to an interior bay of a warehouse. The height of the center column is 15 ft and each beam is 18 ft in length. What is the effective length factor 'K' of the center column?

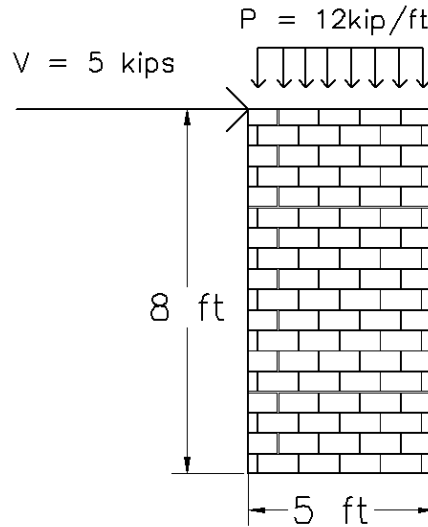
$E = 29000$ ksi
 $F_y = 50$ ksi



- (A) 0.5
- (B) 0.7
- (C) 1
- (D) 1.2

60. A fully grouted masonry wall is subjected to a 12 kip/ft distributed load at the top and a shear of 5 kips acting at the top. The thickness of the wall is 7.625 in and the total area of reinforcement on the both ends is 4 in². The effective cover for the rebars are 4 in. What is the maximum tensile stress acting on the section?

- (A) 0
(B) 0.131 ksi
(C) 0.105 ksi
(D) 0.15 ksi

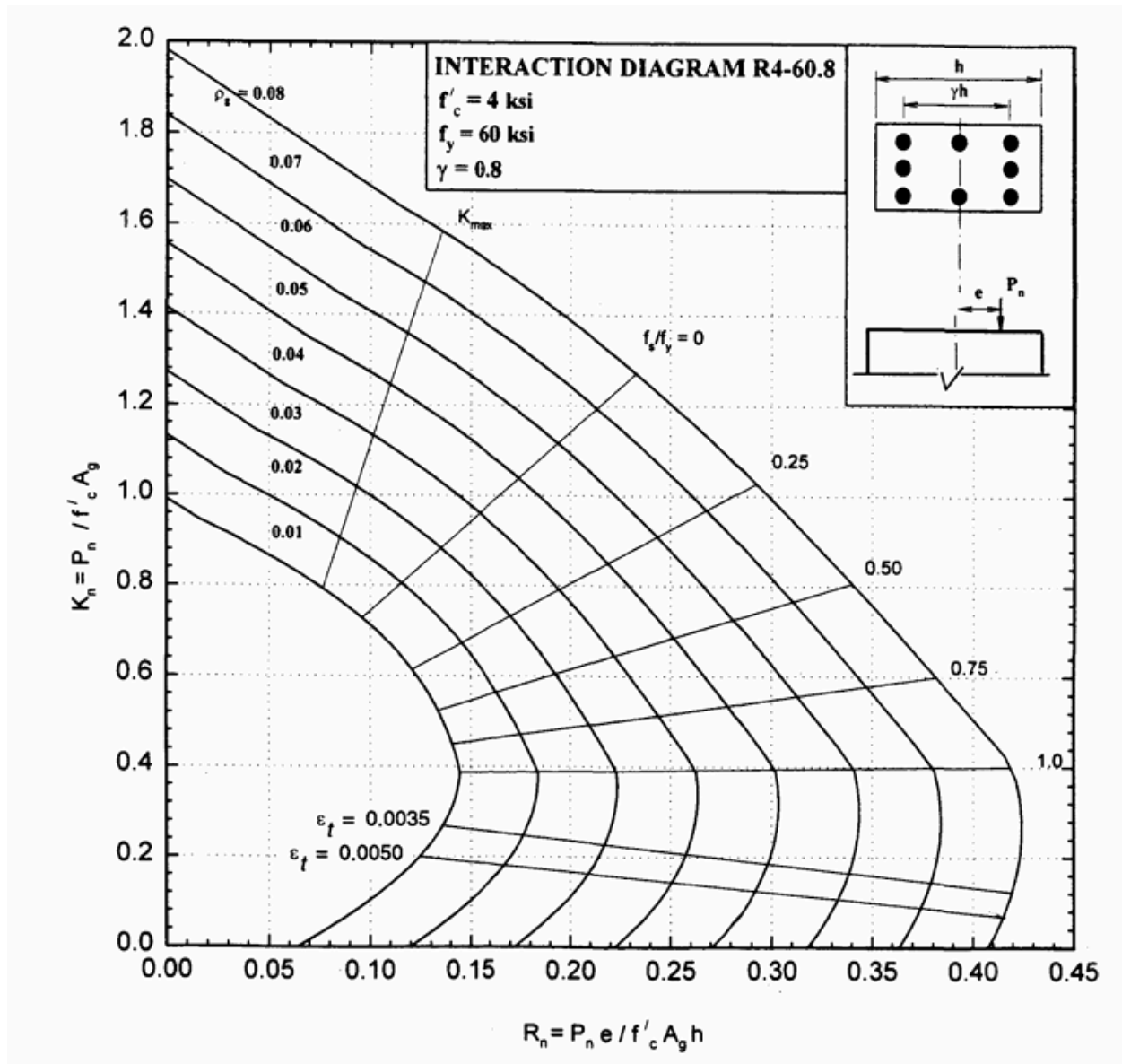


61. What is the nominal axial load capacity of a 16 in X 16 in RCC column section with $f'_c = 6000$ psi? The column has eight # 8 longitudinal reinforcement bars effectively tied per requirements of ACI 318-14. All rebars are Gr 60.

- (A) 859 kips
(B) 1322 kips
(C) 1652 kips
(D) 1943 kips

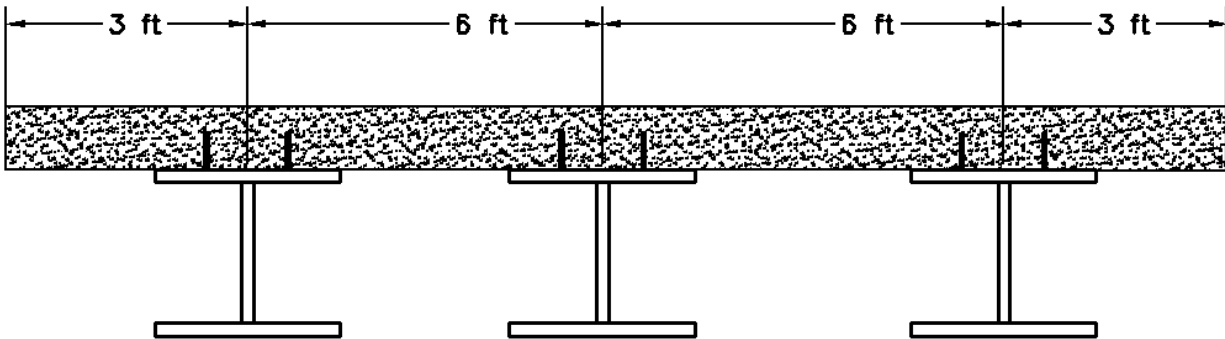
62. A 16 inch X 16 inch square column is subjected to a factored axial load of 798 kips and a factored moment of 66.5 kip*ft. The interaction diagram for the section is shown below. What is the required area of steel for the section?

$$f'_c = 4 \text{ ksi} \quad f_y = 60 \text{ ksi} \quad \Phi = 0.65$$



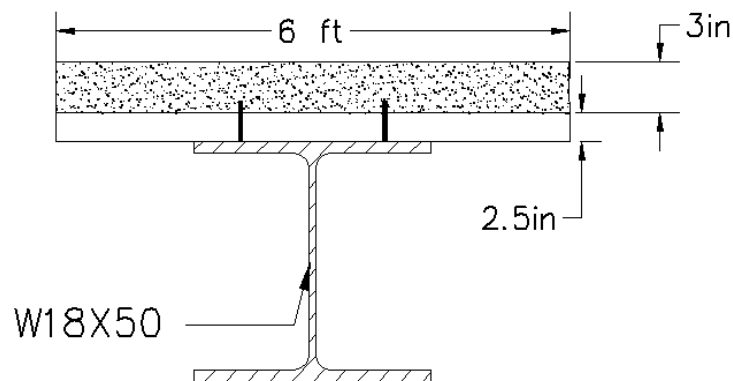
- (A) 3 in²
- (B) 5.5 in²
- (C) 8.6 in²
- (D) 10.5 in²

63. The section of a composite deck system is shown below. What is the total effective width of the concrete slab to be considered for the composite beam on the edge? The center to center distance between supports of the beam along the span is 16ft.



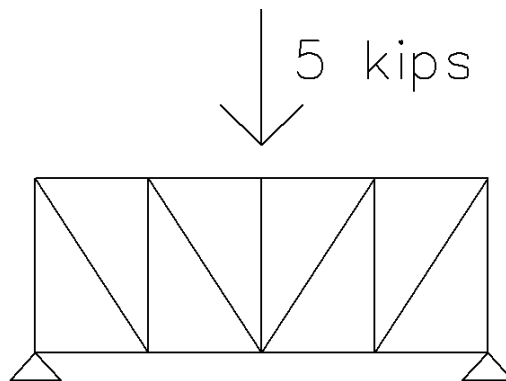
- (A) 2ft
- (B) 4ft
- (C) 6ft
- (D) 8ft

64. A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 64 and the nominal shear strength of each stud is 16.28 kips. What is the allowable moment capacity of the composite section? Strength of concrete slab $f'_c = 4250$ psi

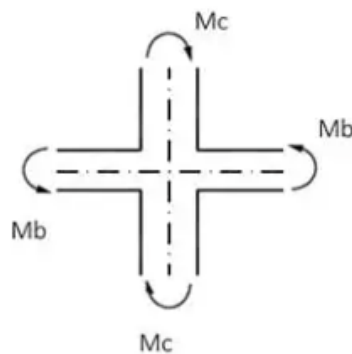


- (A) 319 kip*ft
- (B) 446 kip*ft
- (C) 513 kip*ft
- (D) 544 kip*ft

65. A truss shown below is subjected to a concentrated factored load of 5 kips at the center span. The span of the truss is 20 ft and the height of the truss is 5 ft. What is the maximum design tension force at the bottom chord of the truss?

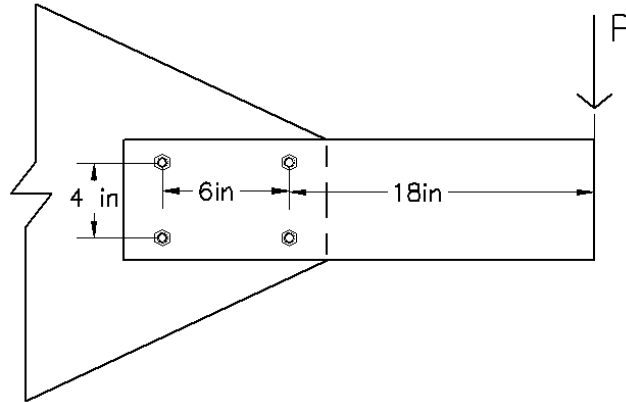


- (A) 4 kips
(B) 5 kips
(C) 6 kips
(D) 7 kips
66. A reinforced concrete special moment frame is to be designed for a building. For a critical joint, the sum nominal flexural strengths of columns framing into it is 1200 kip*ft. What is the maximum permissible sum of nominal flexural strength of beams framing into the joint so that strong -column-weak-beam criteria is satisfied ?

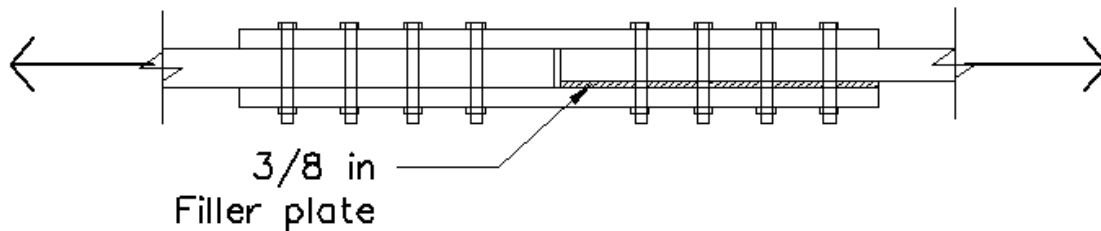


- (A) 600 kip*ft
(B) 800 kip*ft
(C) 1000 kip*ft
(D) 1200 kip*ft

67. The figure below shows a bolted connection that is supporting an eccentric load. The governing failure of the connection is bolt shear strength. If the allowable shear strength of the bolt is 22 kips, what is the maximum load “P” that the connection can support?

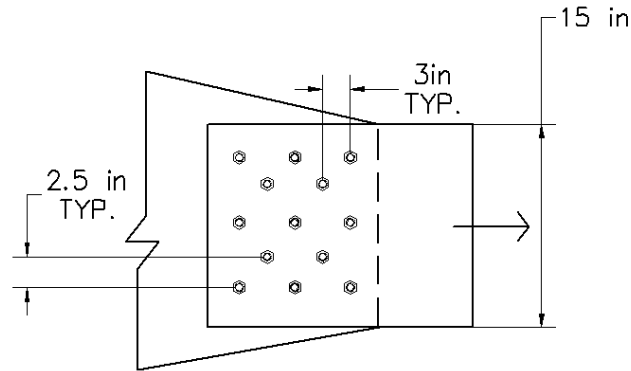


- (A) 13.2 kips
 (B) 22 kips
 (C) 36.7 kips
 (D) 88 kips
68. A tension splice is constructed for two plates with different thickness as shown below. To make up for the gap, a $\frac{3}{8}$ in thick filler plate is used. The fasteners used are $\frac{5}{8}$ inch diameter A307 grade bolts. What is the allowable shear strength of an individual bolt to be used for the connection?

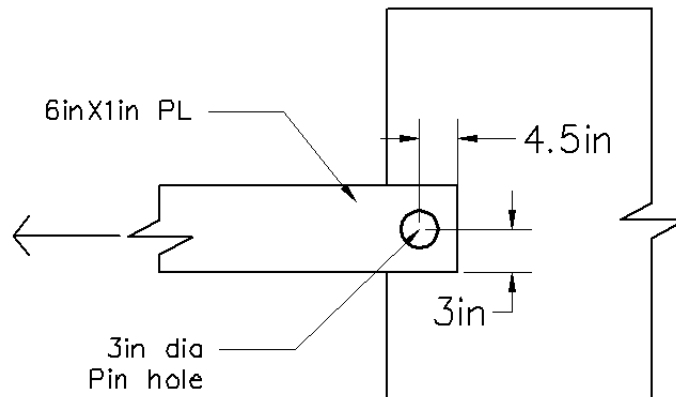


- (A) 4.14 kips
 (B) 7.87 kips
 (C) 8.29 kips
 (D) 12.5 kips

69. Two plates are connected to each other as shown below. The thickness of the thinner top plate is 0.5 in. The nominal size of bolt holes is $\frac{13}{16}$ in. What is the nominal tensile rupture strength of the top plate? $F_u = 58$ ksi

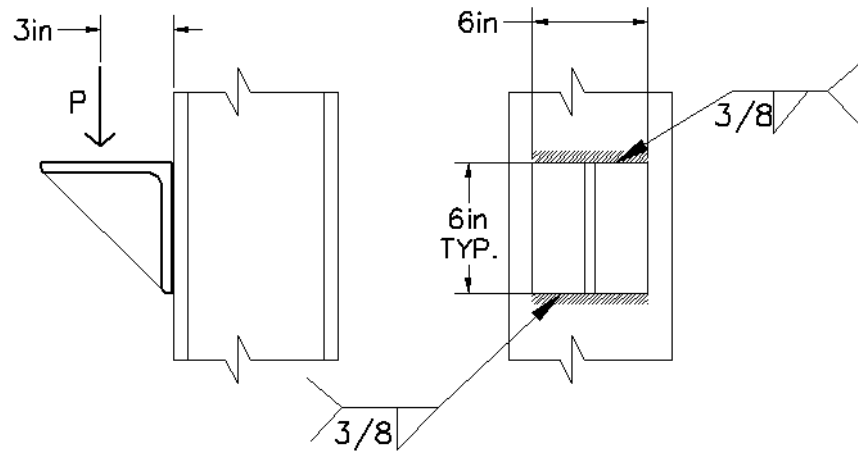


- (A) 269 kips
 (B) 315 kips
 (C) 359 kips
 (D) 390 kips
70. A 6 in. X 1 in. pin plate is connected to a structural member as shown below. The diameter of the pin is 2.875 in. and the hole diameter is 3 in. What is the design strength of the pin plate in tension? Assume pin failure in shear and bearing do not control. $F_y = 36$ ksi and $F_u = 58$ ksi for plates.



- (A) 130.5 kips
 (B) 194.4 kips
 (C) 231.6 kips
 (D) 258.2 kips

71. A bracket is welded on the face of the column as shown below. Using elastic method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection. $F_{exx} = 70 \text{ ksi}$



- (A) 29.5 kips
(B) 38.7 kips
(C) 61.3 kips
(D) 70.9 kips

72. In a bearing type connection, a 1 in. diameter ASTM A325 bolt is subjected to both shear and tension loads (single shear with threads not included in shear plane). The shear load in the bolt is 5 kips. What is the allowable tensile strength of the bolt when both shear and tension are acting together on the bolt?

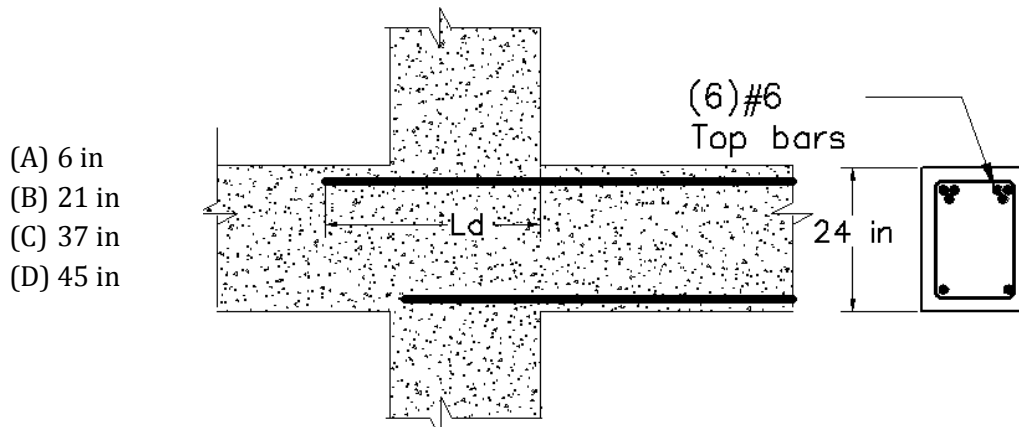
- (A) 24.7 kips
(B) 26.3 kips
(C) 29.1 kips
(D) 35.3 kips

73. A welded connection uses a double-sided fillet weld to connect a plate to a column flange. The weld is designed to resist a shear load, but nondestructive testing reveals a 1-inch-long lack of fusion in one of the welds. Which statement is not true about the impact of this imperfection on the connection's performance?

- (A) The lack of fusion reduces the effective throat, decreasing the weld's shear capacity proportionally,
- (B) The lack of fusion affects the weld's fatigue life and has negligible effect on its static strength,
- (C) The lack of fusion increases the likelihood of brittle fracture under loadings,
- (D) The imperfection acts as a stress concentration point, increasing the likelihood of crack propagation.

74. The beam column connection shown below has bundled #6 Gr. 60 deformed bars (three rebars in one bundle) at the top. What is the development length 'Ld' of the top bars in tension from the face of the column? The clear cover is 2 in and the clear spacing between the bundles is 7 in.

$f'_c = 4000 \text{ psi}$, $\Psi_e = 1$, $\lambda = 1$

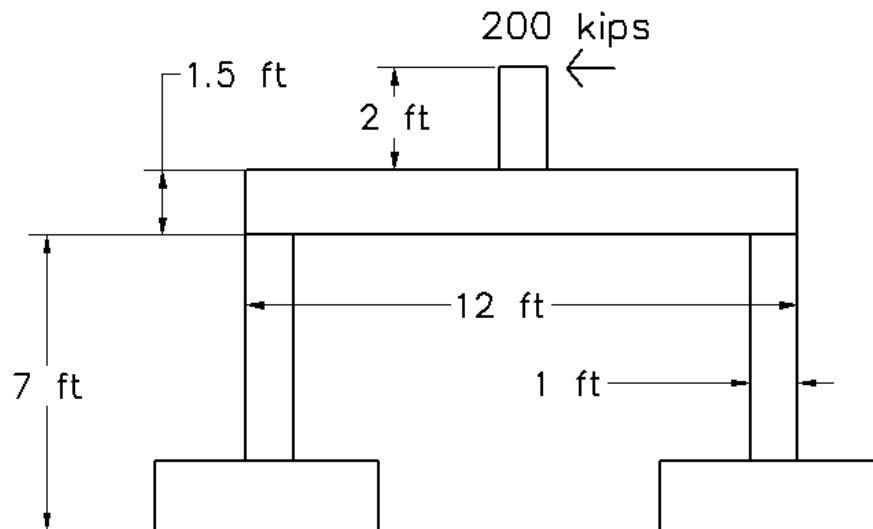


- (A) 6 in
- (B) 21 in
- (C) 37 in
- (D) 45 in

75. A 6 ft X 4 ft shallow footing is proposed at 2 ft below grade in a region with a frost depth of 3.5 ft. The soil is sandy with an allowable bearing pressure of 2.5 ksf. What is the most appropriate measure to ensure frost protection?

- (A) The footing depth can be kept at 2 ft and soil should be insulated from thermal effects,
- (B) The footing depth has to be increased to 3.5 ft or deeper.
- (C) The footing size has to be reduced to limit frost heaving,
- (D) The foundation design has to be revised to pile foundations with sufficient capacity.

76. The figure below shows two columns supported by 4 ft X 4 ft square footings. The vertical modulus of subgrade reaction of the bearing strata is 150 lb/in³. What is most nearly the settlement under the footing due to the lateral load of 200 kips. Self weight of the structure can be neglected and the stiffness of the structure can be assumed to be very high.

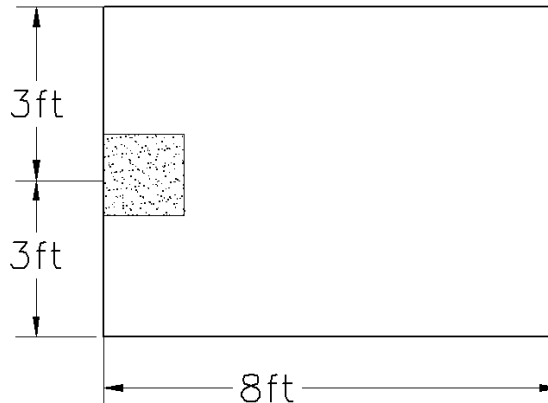


- (A) 0.24 in
- (B) 0.38 in
- (C) 0.55 in
- (D) 0.67 in

77. The plan of a RCC 6ftX8ft footing is shown below. The footing has an eccentric column at the edge. The dimensions of the column is 18inX18in. What is the design bearing capacity of the column on the footing if the effective depth of the footing is 21 in?

$$f'_c = 4 \text{ ksi} \quad \lambda = 1$$

- (A) 612 kips
(B) 716 kips
(C) 941 kips
(D) 1100 kips

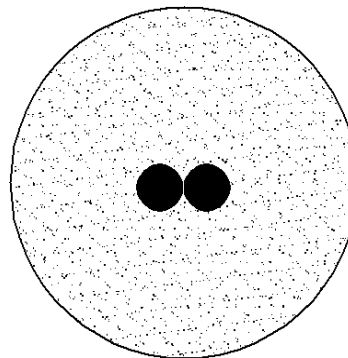


78. Which of the following is true about end bearing piles and friction piles?

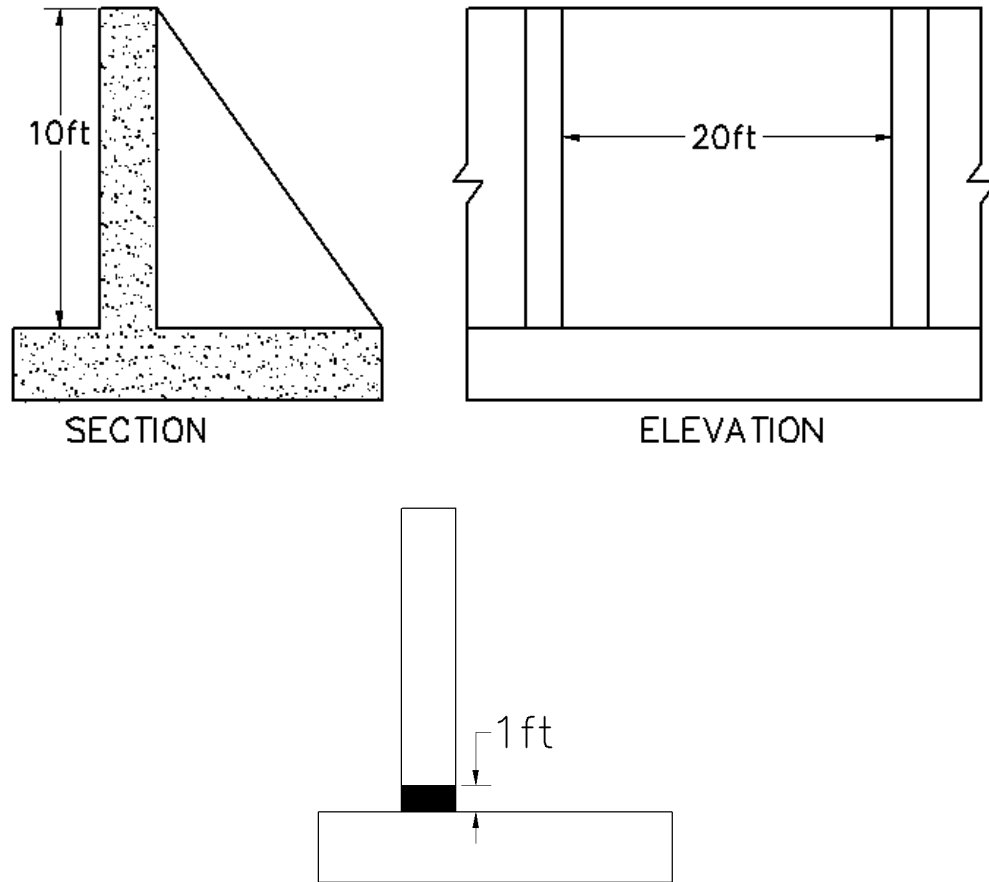
- (A) End bearing piles are shorter than friction piles,
(B) The bearing capacity of end bearing piles depends upon pile material whereas the bearing capacity of skin friction piles are only affected by the properties of surrounding soil,
(C) The phenomenon of negative skin friction can affect both end bearing piles and friction piles,
(D) End bearing piles are used in cohesionless soil whereas friction piles are used in cohesive soil.

79. The figure below shows the cross section of a reinforced concrete micropile that is cast in a rock socket. The diameter of the micropile is 12 in. The compressive strength of grout is $f'_c = 6000$ psi. The two center rebars used are #18 Grade 75. What is the allowable compression in the micropile? Area of #18 bar = 4 in^2

- (A) 328 kips
(B) 368 kips
(C) 402 kips
(D) 448 kips



80. The interior segment of a 100 ft long counterfort RCC retaining wall system is shown below. The wall is retaining backfill on the counterfort side. The height of the backfill is equal to the height of the wall. What is most nearly the maximum factored moment to be considered for the design of horizontal flexural reinforcement for the bottom 1 ft strip of the stem? Density of soil = 120 pcf, Active earth pressure coefficient, $K_a = 0.33$



- (A) 25 kip*ft
- (B) 30 kip*ft
- (C) 35 kip*ft
- (D) 40 kip*ft

Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (D) 250 tons

Explanation: {Ref NCEES manual page 3 for conversion factors}

Convert kN to lbf

$$1\text{ N} = 0.225\text{ lbf}$$

$$1\text{ kN} = 1000 * 0.225\text{ lbf}$$

$$1\text{ kN} = 225\text{ lbf}$$

$$2020\text{ kN} = 2020 * 225\text{ lbf} = 454500\text{ lbf}$$

Convert lbf to (short) ton

$$2000\text{ lbf} = 1\text{ ton}$$

$$454500\text{ lbf} = (1/2000) * 454500\text{ tons}$$

$$= 227.25\text{ tons}$$

Thus a jack with capacity greater than 227.25 tons is required. Use **250 tons** jack from the options.

2. Problem

Correct Answer: (B) 468 lb

Explanation:

{Ref ASCE 7-16 page 430 Table C3.1-2 for unit weight of brick masonry}

$$\text{Unit Weight of hard brick masonry} = 130\text{ lb/ft}^3$$

$$\text{Area of masonry wall} = 9\text{ inches} * 4\text{ft} = 3\text{ft}^2$$

$$\begin{aligned}\text{Uniformly distributed dead load due to wall} &= 130\text{ lb/ft}^3 * 3\text{ft}^2 \\ &= 390\text{ lb/ft}\end{aligned}$$

$$\text{Factored dead load to be used in combination with live load} = 1.2 * 390 = \mathbf{468\text{ lb/ft}}$$

3. Problem

Correct Answer: (C) 34 kips

Explanation:

{Ref ASCE 7-16 page 16 and 17 for live load reduction provisions}

$K_{LL} = 4$ Live load element factor From Table 4.7-1

$L_o = 57.5$ kips

$A_T = 500$ ft²

Calculate reduced live load using equation 4.7-1

$$L = L_o \left(0.25 + \frac{15}{\sqrt{K_{LL} A_T}} \right)$$

$$L = 57.5 \text{ kips} * \left(0.25 + \frac{15}{\sqrt{4 * 500}} \right)$$

$$L = 57.5 \text{ kips} * 0.585 \text{ (Check reduction should not be less than } 0.4 L_o, \text{ OK)}$$

$$L = 33.63 \text{ kips} \approx \mathbf{34 \text{ kips}}$$

4. Problem

Correct Answer: (B) ASCE 7-16 specified concentrated live loads are summed with minimum uniformly distributed live loads to create the most severe effect in the structural member

Explanation:

Per ASCE 7-16 page 13 section 4.4, concentrated loads are assumed to act uniformly over an area of 762mmX762mm = 0.58 m².

ASCE 7-16 does not provide provisions for reductions of concentrated live loads.

Concentrated loads govern punching and flexural failure of slabs and are usually more critical in slab design than uniform loads.

All other options are correct except that minimum concentrated loads and minimum uniform loads given in the code are summed. They are not summed. They are checked separately to check which condition governs the design.

5. Problem

Correct Answer: (D) Partially open

Explanation: Ref ASCE 7-16 page 245 for openings definitions

Calculate total area of building envelope

$$\text{Area of walls} = 4 \times 20\text{ft} \times 8\text{ft} = 640 \text{ ft}^2$$

$$\text{Area of roof} = 20\text{ft} \times 20\text{ft} = 400 \text{ ft}^2$$

Check condition for enclosed building

$$\text{Area of opening 01} = 2\text{ft} \times 3\text{ft} = 6 \text{ ft}^2$$

$$\text{Gross area of wall} = 8\text{ft} \times 20\text{ft} = 160 \text{ ft}^2$$

Since area of opening is greater than $0.01 \times \text{Area of wall}$, it is not enclosed.

Check condition for open building

$$\text{Area of openings in front wall 01+02} = A_0 = 2\text{ft} \times 3\text{ft} + 2\text{ft} \times 6\text{ft} = 18 \text{ ft}^2$$

This is less than 80% of wall area. Thus building is not open

Check condition for partially enclosed.

$$\text{Total area of openings excluding 01 and 02} = A_{01} = 03+04+05$$

$$= 2\text{ft} \times 2\text{ft} + 2\text{ft} \times 2\text{ft} + 4\text{ft} \times 4\text{ft} = 24 \text{ ft}^2$$

$$\text{Check } A_0 > 1.1 \times A_{01}, 18 \text{ ft}^2 \text{ is not greater than } 1.1 \times 24 \text{ ft}^2$$

Hence the shed does not qualify as partially enclosed.

Thus correct answer is partially open

6. Problem

Correct Answer: (A) 9.25 psf, 15.91 psf

Explanation:

From ASCE 7-16, chapter 26, Table 26.10-1 for 10ft, Exposure C, Velocity pressure coefficient $K_z = 0.85$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for MWFRS

$$K_{zt} = 1 \quad K_e = 1$$

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2 \quad (\text{equation 26.10-1})$$

$$q_z = 0.00256 * 0.85 * 1 * 0.85 * 1 * 100^2 = 18.5 \text{ psf}$$

From page 276, Figure 27.3-1, for windward wall, for all values of L/B, $C_p = 0.8$

From page 271, Table 26.13-1, for enclosed buildings, value of internal coefficient (GC_{pi}) = +0.18 and -0.18

From page 269, section 26.11.1, gust effect factor $G = 0.85$

From page 274, equation 27.3-1,

Design pressure

$$p = q * G * C_p - q_i(GC_{pi}) \quad q_i = q_z = 18.5 \text{ psf}$$

Calculate design pressure for two internal pressure coefficients

$$p_1 = 18.5 * 0.85 * 0.8 - 18.5 * (0.18) = 9.25 \text{ psf}$$

$$p_2 = 18.5 * 0.85 * 0.8 - 18.5 * (-0.18) = 15.91 \text{ psf}$$

Correct answer is option A both positive (thrust pressure)

7. Problem

Correct Answer: (B) 150 kips

Explanation:

Given

Weight $W = 600$ kips

Mapped Short period spectral acceleration $S_s = 0.5g$

From ASCE 7-16, page 119, Section 12.14.8.1, for simplified procedure, for rocky sites, $F_a = 1$

$$S_{DS} = (2/3) * F_a * S_s$$

$$= 0.667 * 1 * 0.5g = 0.333g$$

Response modification factor for simplified procedure is given by Table 12.14-1, page 115

For ordinary plain masonry walls, $R = 1.5$

For two story building, from page 119 $F = 1.1$

Then, seismic base shear is given by equation 12.14-12

$$V = \frac{F * S_{DS}}{R} * W$$

$$V = \frac{1.1 * 0.333}{1.5} * 600 \text{ kips} = 146.5 \text{ kip} \approx \mathbf{150 \text{ kips}}$$

8. Problem

Correct Answer: (C) 80 psi

Explanation:

Ref AASHTO LRFD specs 2018 page 3-23, section 3.6.1.2.5 for tire contact area

$$A = 10\text{in} * 20\text{in} = 200\text{in}^2$$

Given, maximum weight of rear axle = 32 kip

This load is split in two sides

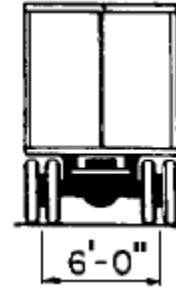
Total wheel load on each side

$$P = 32\text{ kips} / 2 = 16\text{ kips}$$

Pressure = Load/ area

$$= 16\text{ kips} / 200\text{in}^2$$

$$= 16000\text{ lb} / 200\text{in}^2 = \mathbf{80\text{ psi}}$$



9. Problem

Correct Answer: (B) 46.2 psf

Explanation:

From ASCE 7-16, page 266, section 26.7.3, for urban and semi urban area, surface roughness and exposure is category B

From ASCE 7-16, Table 1.5-2, page 5, for risk category III,
snow importance factor $I_s = 1.1$

From page 58, Table 7.3-1, for surface roughness B and partially exposed roof,
exposure factor $C_e = 1.0$

From page 58, Table 7.3-2, for unheated roof,
thermal factor $C_t = 1.2$

For anchorage Alaska, from Table 7.2-1 page 54,

Ground snow load, $p_g = 50\text{psf}$

From equation 7.3-1 page 51, flat roof snow load

$$p_f = 0.7 * C_e * C_t * I_s * p_g$$

$$p_f = 0.7 * 1 * 1.2 * 1.1 * 50\text{psf} = \mathbf{46.2\text{ psf}}$$

10. Problem

Correct Answer: (B)

Explanation:

All options are correct except option B. Both the weight of ice and wind on ice are considered together and are in the same load combination as given in ASCE 7-16 page 7, section 2.3.3. They are not considered separately.

11. Problem

Correct Answer: (C) 120 kips

Explanation:

Calculate soil lateral loads per IBC 2018 section 1610 page 391.

Since water table is below the foundation mat, hydrostatic pressures don't apply.

The foundation wall is extended up to the first floor slab and can't freely move at the top.

Hence, the pressure that applies is earth pressure at rest

From page 392 table 1610.1 for well graded gravel-sand mixes,

At-rest pressure = 60psf/ ft depth.

For a 10feet deep wall, the maximum intensity at the bottom of the wall

= 60psf/ft depth * 10ft = 600psf

Soil pressure varies linearly from 0 at the top of wall to maximum intensity at the bottom.

Hence average lateral load along the vertical face of wall = $\frac{1}{2}$ *max intensity* height

= $\frac{1}{2}$ * 600 psf * 10 ft = 3000 lb/ft = 3kips/ft

Thus, total lateral load on a wall of 40 ft length = 3kip/ft * 40 ft = **120 kips**

12. Problem

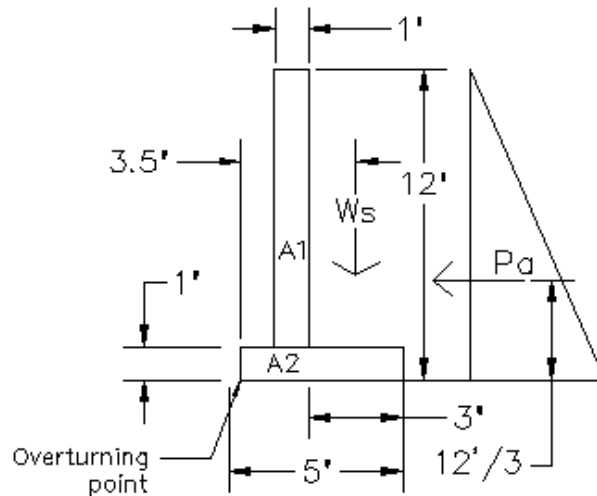
Correct Answer: (C) 1.6

Explanation:

Active Earth pressure coefficient, $K = 0.33$

Density of soil, $\gamma = 115$ pcf

Density of concrete, $\gamma = 150$ pcf



The forces resisting overturning are the self weight of the wall and the weight of soil retained by the toe of the wall.

Weight of soil retained by wall

$$W_s = (12\text{ft} - 1\text{ft}) * 3\text{ft} * 115\text{pcf}$$

$$= 3795 \text{ lb/ft}$$

$$\text{Distance of centroid of retained soil weight to overturning point} = (5\text{ft} - 3\text{ft} / 2) = 3.5\text{ft}$$

Weight of wall is broken into A1 and A2

$$\text{Weight of stem A1} = (12\text{ft} - 1\text{ft}) * 1\text{ft} * 150\text{pcf} = 1650 \text{ lb/ft}$$

$$\text{Distance of centroid of A1 to overturning point} = (5\text{ft} - 3\text{ft} - 1\text{ft} / 2) = 1.5\text{ft}$$

$$\text{Weight of base A2} = 5\text{ft} * 1\text{ft} * 150\text{pcf} = 750 \text{ lb/ft}$$

$$\text{Distance of centroid of A2 to overturning point} = 5\text{ft} / 2 = 2.5\text{ft}$$

$$\text{Resisting moment} = W_s * 3.5\text{ft} + A1 * 1.5\text{ft} + A2 * 2.5\text{ft}$$

$$= 3795 * 3.5 + 1650 * 1.5 + 750 * 2.5$$

$$= 17632.5 \text{ lb*ft/ft}$$

$$\text{Overturning force, active earth pressure} = \frac{1}{2} * K_a * \gamma * H^2$$

$$= 0.5 * 0.33 * 115 * 12\text{ft} * 12\text{ft} = 2732.4 \text{ lb/ft}$$

$$\text{Overturning moment arm} = 12\text{ft} / 3 = 4\text{ft}$$

$$\text{Overturning moment} = 2732.4\text{lb/ft} * 4\text{ft} = 10929.6 \text{ lb*ft/ft}$$

$$\text{Factor of safety against overturning} = \text{Overturning moment} / \text{resisting moment}$$

$$= (17632.5 \text{ lb*ft/ft}) / 10929.6 \text{ lb*ft/ft} = \mathbf{1.6}$$

13. Problem

Correct Answer: (C) 26 kips

Explanation:

All walls have equal in plane stiffness.

Let stiffness be 'k'

The center of rigidity is at (20ft/2, 12ft/2) from the bottom left corner.

Total lateral load

$$= 3 \text{ kip/ft} \times 12 \text{ ft} = 36 \text{ kips}$$

Total torsional moment

$$M = 360 \text{ kip}\cdot\text{ft}$$

Direct shear on North and south walls

$$V = 36 \text{ kips} \cdot k / (k + k)$$

$$V = 18 \text{ kips}$$

Calculate torsional shear.

Distance of walls from center of rigidity

$$r_1 = r_2 = 6 \text{ ft for north and south walls}$$

$$r_3 = r_4 = 10 \text{ ft for east and west wall}$$

Max shear is in north wall with combined direct and torsional shear

$$V_s = \frac{M \cdot k_i \cdot r_i}{\sum k_i \cdot r_i^2}$$

$$V_s = \frac{360 \text{ kip}\cdot\text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 10 \text{ ft} \cdot 10 \text{ ft} + k \cdot 10 \text{ ft} \cdot 10 \text{ ft} + k \cdot 6 \text{ ft} \cdot 6 \text{ ft} + k \cdot 6 \text{ ft} \cdot 6 \text{ ft}}$$

$$V_s = \frac{360 \text{ kip}\cdot\text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 272 \text{ ft}^2}$$

$$V_s = \frac{360 \text{ kip}\cdot\text{ft} \cdot k \cdot 6 \text{ ft}}{k \cdot 272 \text{ ft}^2}$$

$$V_s = 7.94 \text{ kips}$$

$$\text{Total shear} = V + V_s = 18 \text{ kips} + 7.94 \text{ kips} = 25.94 \text{ kips} \approx \mathbf{26 \text{ kips}}$$

14. Problem

Correct Answer: (C) **24kip*ft/ft**

Explanation:

Moment due to earth load, $EH = 10 \text{ kip*ft/ft}$

Moment due to live load surcharge, $LS = 5.5 \text{ kip*ft/ft}$

Moment due to wind load on structure, $WS = 1.5 \text{ kip*ft/ft}$

From AASHTO LRFD 2017 page 3-15,

Factor for $LS = 1.35$

Factor for $WS = 1$

From page 3-16, Factor for $EH = 1.5$

Thus, max factored moment $= 1.5EH + 1.35LS + 1WS$

$= 1.5 * 10 + 1.35 * 5.5 + 1 * 1.5 = 23.92 \text{ kip*ft/ft} \approx \mathbf{24 \text{kip*ft/ft}}$

15. Problem

Correct Answer: (C)

Explanation:

An external couple does not produce any abrupt change in the shear force diagram.

Hence Options B and D are ruled out.

The shear force diagram, when there is a uniformly distributed load should be linear. This rules out option A. Hence the correct answer is Option C.

16. Problem

Correct Answer: (B)

Explanation:

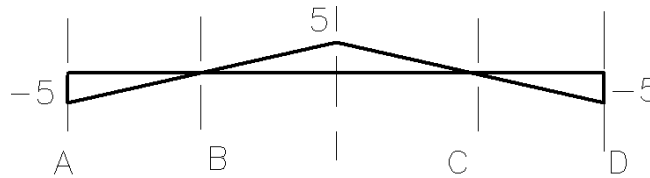
All options are correct except option B. The slope of the shear force diagram at any point gives the maximum intensity of loading at that point, not the moment.

17. Problem

Correct Answer: (B)

Explanation:

At midspan, the moment is negative (tension at top) when the load travels along the overhangs and positive when it travels between the supports. The maximum moment at the midspan occurs when the load is directly on top of it. The value of ordinate at that point is $(a*b/L)$, where 'a' is distance to left support from midspan = 10ft, 'b' is the distance to right support from midspan = 10 ft and 'L' is the span length between supports = 20ft. The overhang diagrams are then linearly extrapolated on both sides



The correct answer is Option B.

18. Problem

Correct Answer: (C) Catenary curve

Explanation:

Chains, ropes and cables assume a catenary shape when they are hanging under their own weight. Under uniformly distributed external load, the shape will be parabolic, Hence the correct answer is option C

19. Problem

Correct Answer: (A) 378.5 kip/in

Explanation:

Here, when vertical load acts, the deformation in each braces will be proportional to the individual stiffness. Axial force acting in each layer will be equal to the total force and the deformation will be the sum of deformations in each layer. Thus the springs are in series and the total stiffness will be

$$1/K = 1/K_1 + 1/K_2 + 1/K_3$$

For first layer,

$$E1 = 200 \text{ ksi} \quad A1 = 20 \text{ in}^2 \quad L1 = 4 \text{ in}$$

$$K1 = E1 \cdot A1 / L1 = 200 \text{ ksi} \cdot 20 \text{ in}^2 / 4 \text{ in} = 1000 \text{ kip/in}$$

For second layer,

$$E2 = 300 \text{ ksi} \quad A2 = 16 \text{ in}^2 \quad L2 = 5 \text{ in}$$

$$K2 = E2 \cdot A2 / L2 = 300 \text{ ksi} \cdot 16 \text{ in}^2 / 5 \text{ in} = 960 \text{ kip/in}$$

For second layer,

$$E3 = 250 \text{ ksi} \quad A3 = 20 \text{ in}^2 \quad L3 = 3 \text{ in}$$

$$K3 = E3 \cdot A3 / L3 = 250 \text{ ksi} \cdot 20 \text{ in}^2 / 3 \text{ in} = 1666.67 \text{ kip/in}$$

Now,

$$1/K = 1/K1 + 1/K2 + 1/K3 = 1/1000 + 1/960 + 1/1666.67 = 0.00264166 \text{ in/kip}$$

$$\text{Thus, } K = 1/0.00264166 \text{ in/kip} = \mathbf{378.5 \text{ kip/in}}$$

Correct answer is option A.

20. Problem

Correct Answer: (C)4

Explanation:

Member AG is a zero force member. This is because when we analyze joint A, the roller support does not produce any horizontal reaction and AG is the only horizontal member at that point. Thus force in AG = 0

When we analyze joint E, there is no external load. If there are ONLY two non-collinear members at a joint, with no external load, both are zero force members. Hence both ED and EF are zero force members.

When we analyze joint C, there is no external load. If there are ONLY three members at a joint, two of which are collinear, with no external load, then the non collinear member is zero force member. So CG is also a zero force member.

Thus AG, ED, EF and CG are zero force members. So the correct answer is Option C.

21. Problem

Correct Answer: (B) 60 kips

Explanation:

Per ACI 318-14 section 18.4.3 page 271, for intermediate moment resisting frames, the columns design shear should be the least between the shear associated with developing nominal flexural strengths and the factored shear combination considering overstrength factor of 3 for EQ loads.

Factored Dead load shear = $1.2 \times 5 \text{ kips} = 6 \text{ kips}$

Factored Live load shear = $1 \times 10 \text{ kips} = 10 \text{ kips}$

Factored EQ load shear = $3 \times 20 \text{ kips} = 60 \text{ kips}$

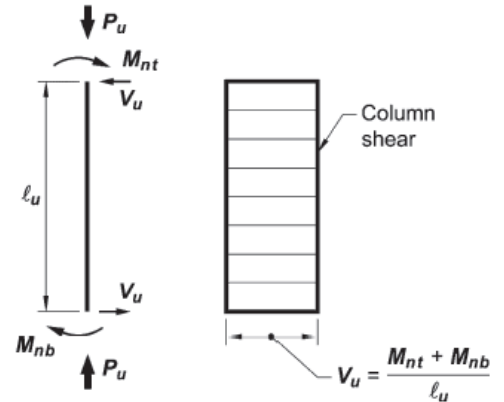
Total factored shear combination = $6 + 10 + 60 = 76 \text{ kips}$

$M_{nt} = 450 \text{ kip}\cdot\text{ft}$ $M_{nb} = 600 \text{ kip}\cdot\text{ft}$

Clear span, $\ell_u = 20 \text{ ft} - 24 \text{ in} = 18 \text{ ft}$

Flexural shear = $(M_{nt} + M_{nb})/\ell_u = (450 \text{ kip}\cdot\text{ft} + 600 \text{ kip}\cdot\text{ft})/18 \text{ ft}$

$= 58.33 \approx 60 \text{ kips}$ Least between the two shears is **60 kips**, thus Option B is correct.



22. Problem

Correct Answer: (A) -6 kips

Explanation:

Ref to NCEES PE handbook page 32.

Moment, $M = x^3 - 5x^2 - 3x$

The shear force is given by the first derivative of the bending moment.

$$V = \frac{d(Mx)}{dx}$$

$$V = \frac{d(x^3 - 5x^2 - 3x)}{dx}$$

$$V = \frac{d(x^3 - 5x^2 - 3x)}{dx}$$

$$V = 3x^2 - 10x - 3$$

When $x = 3 \text{ ft}$

$$V = 3 \times 3^2 - 10 \times 3 - 3 = -6 \text{ kips}$$

Thus, the correct answer is Option A.

23. Problem

Correct Answer: (C) 9.67 kips

Explanation:

The maximum shear will occur when the heaviest axle travels past the maximum reaction point i.e. when the heavier rear axle is at the pinned support location and the front axle is at 8 ft from the pinned support.

Maximum shear

$$V = 8 \text{ kips} + 3 \text{ kips} * (18\text{ft}-8\text{ft})/18\text{ft}$$

= 9.67 kips Thus, the correct answer is option C.

24. Problem

Correct Answer: (A) 23.2 ksi

Explanation:

The steel section is converted to equivalent wood section using modular ratio, n

$$n = (E \text{ steel}) / (E \text{ wood})$$

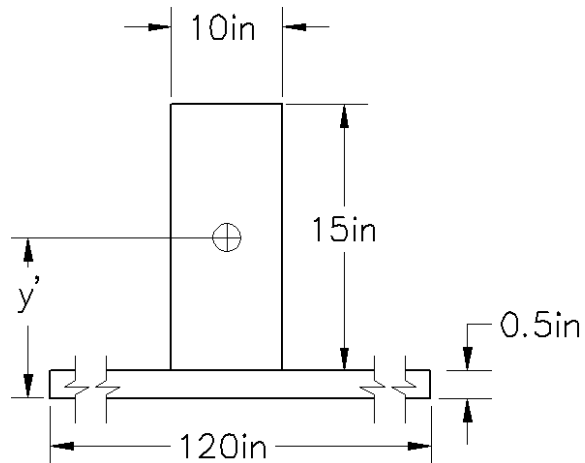
$$= 30000 / 2500 = 12$$

Thus, width of transformed steel plate

$$= 10 \text{ in} * n$$

$$= 10 \text{ in} * 12 = 120 \text{ in}$$

Calculate the centroid and MOI of the transformed section,



Centroid from the base,

$$y' = \{120 * 0.5 * 0.25 + 15 * 10 * (7.5 + 0.5)\} / (120 * 0.5 + 15 * 10)$$

$$Y' = 5.786 \text{ in}$$

Calculate moment of inertia about centroid

$$I_a = 10 * 15 * 15 * 15 / 12 = 2812.5 \text{ in}^4$$

$$I_b = 120 * 0.5 * 0.5 * 0.5 / 12 = 1.25 \text{ in}^4$$

$$I = 2812.5 \text{ in}^4 + 10 \text{ in} * 15 \text{ in} * (8 \text{ in} - y')^2 + 1.25 \text{ in}^4 + 120 \text{ in} * 0.5 \text{ in} * (y' - 0.25 \text{ in})^2$$

$$I = 5388 \text{ in}^4$$

Section modulus for steel fiber

$$S = I/y'$$

$$= 5388 \text{ in}^4 / 5.786 \text{ in}$$

$$= 931.21 \text{ in}^3$$

Flexural stress in steel is given by

$$\sigma = n * M/s$$

$$= \{12 * 150 \text{ kip*ft} * 12 \text{ (in/ft)}\} / 931.21 \text{ in}^3$$

$$= 23.2 \text{ ksi}$$

Thus the correct answer is option A.

25. Problem

Correct Answer: (D)

Explanation:

The shape factor of I beams is around 1.14 and of rectangular cross sections is 1.5.

Structures using plastic moment capacity are generally economical. PNA divides areas of stress blocks into two equal halves. However, for hybrid sections, the plastic section modulus is different from that of a homogenous section with the same shape and dimensions. So Option D is false and the answer.

26. Problem

Correct Answer: (B) 6.07 kip*ft

Explanation:

Ref NCEES PE handbook page 232

Let 'I' be the uniform moment of inertia,

Let 'E' be the uniform modulus of elasticity,

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness "k" for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

Calculate the stiffness of each member accordingly.

$$K_{oa} = 4EI/9 = 0.4444 EI$$

$$K_{ob} = 3EI/11 = 0.2727 EI \text{ (far end hinged)}$$

$$K_{oc} = 3EI/13 = 0.2307 EI \text{ (far end hinged)}$$

$$K_{od} = 4EI/8 = 0.5 EI$$

$$K_{oe} = 4EI/5 = 0.8EI$$

Total joint stiffness

$$K = K_{oa} + K_{ob} + K_{oc} + K_{od} + K_{oe}$$

$$= 0.4444 EI + 0.2727 EI + 0.2307 EI + 0.5 EI + 0.8 EI$$

$$= 2.2478 EI$$

Balancing moment distributed to OB

$$= k_{ob}/K * 50 \text{ kip}\cdot\text{ft}$$

$$= 0.2727EI/2.2478 EI * 50 \text{ kip}\cdot\text{ft} = \mathbf{6.07 \text{ kip}\cdot\text{ft}}$$
 Correct answer is option B.

27. Problem

Correct Answer: (D) 2175 psi compressive

Explanation:

$$\text{Area, } A = 240 \text{ in}^2$$

$$\text{Section Modulus, } S = 800 \text{ in}^3$$

$$\text{Prestress, } P = 90 \text{ kips}$$

$$\text{eccentricity, } e = -4 \text{ in}$$

Axial compression due to prestress

$$f_1 = P/A$$

$$= -90000 \text{ lbs} / 240 \text{ in}^2 = -375 \text{ psi}$$

$$\text{Moment due to prestress eccentricity, } M_p = P \cdot e$$

$$= 90000 \text{ lbs} * 4 \text{ in} = 360000 \text{ lbs}\cdot\text{in}$$

Since Prestress is applied below the centroid, eccentricity will cause tension at the top,

$$f_2 = M_p / S$$

$$= 360000 \text{ lbs}\cdot\text{in} / 800 \text{ in}^3 = 450 \text{ psi}$$

$$\text{Moment due to UDL, } M_{udl} = wL^2/8$$

$$= 3 \text{ kip/ft} * 20\text{ft} * 20\text{ft} / 8$$

$$= 150 \text{ kip}\cdot\text{ft} = 1800000 \text{ lbs}\cdot\text{in}$$

This bending will cause compression at the top.

$$f_3 = M_{udl}/S$$

$$= -1800000 \text{ lbs}\cdot\text{in} / 800 \text{ in}^3 = -2250 \text{ psi}$$

$$\text{Total stress at the top} = f_1 + f_2 + f_3$$

$$= -375 \text{ psi} + 450 \text{ psi} - 2250 \text{ psi}$$

$$= -2175 \text{ psi} = \mathbf{2175 \text{ psi compressive}}$$

Correct answer is option D

28. Problem

Correct Answer: (B) 18 kip*ft

Explanation:

Maximum bending moment acting in the shaft,

$$M = 15 \text{ kip*ft}$$

Maximum twisting moment acting in the shaft

$$T = 10 \text{ kip*ft}$$

The equivalent twisting moment that produces the same maximum shear as both T and M is

$$T_{eq} = \sqrt{T^2 + M^2}$$

$$T_{eq} = \sqrt{(10 \text{ kip} * \text{ft})^2 + (15 \text{ kip} * \text{ft})^2} = \mathbf{18 \text{ kip*ft}}$$

Correct answer is option B.

29. Problem

Correct Answer: (C) 18.76 ksi

Explanation:

Ref NCEES PE handbook page 28-29 for stresses, strains and thermal deformations,

$$\text{Area, } A = 11 \text{ in}^2$$

$$\text{Length, } L = 20 \text{ ft}$$

$$\alpha = 0.000007/^{\circ}\text{F}$$

$$E = 29000 \text{ ksi}$$

$$\Delta T = 102 - 32 = 70^{\circ}\text{F}$$

Change in length due to load ' ΔT '

$$\Delta L = \alpha * L * \Delta T$$

$$\text{Thermal strain} = \Delta L / L = \alpha * \Delta T$$

$$\text{Thermal stress} = \text{Thermal strain} * E = \alpha * \Delta T * E$$

$$= 0.000007 * 70 * 29000 \text{ ksi}$$

$$= 14.21 \text{ ksi (compressive)}$$

The axial load acts at the center of the span. Hence, the reaction is equal on both ends.

$$R = 100 \text{ kips} / 2 = 50 \text{ kips}$$

$$\text{Axial stress due to compressive reaction on right end} = R / A$$

$$= 50 \text{ kips} / 11 \text{ in}^2 = 4.55 \text{ ksi}$$

$$\text{Total compressive stress} = 14.21 \text{ ksi} + 4.55 \text{ ksi} = \mathbf{18.76 \text{ ksi}} \quad \text{Correct answer is option C .}$$

30. Problem

Correct Answer: (B) 2.27in

Explanation:

Ref ASCE 7-16 page 103, section 12.8.6 to calculate deflection associated with story drift

$$\delta_x = (C_d \cdot \delta_e) / I_e$$

Deflection amplification factor from page 91 ,

$$C_d = 5.5$$

For risk category IV, seismic importance factor From Table 1.2-2 page 5,

$$I = 1.5$$

$$\delta_5 = 5.5 \cdot 2.1 / 1.5 = 7.7 \text{ in}$$

$$\delta_4 = 5.5 \cdot 1.63 / 1.5 = 5.98 \text{ in}$$

$$\delta_3 = 5.5 \cdot 1.14 / 1.5 = 4.18 \text{ in}$$

$$\delta_2 = 5.5 \cdot 0.62 / 1.5 = 2.27 \text{ in}$$

$$\delta_1 = 0$$

Maximum difference in displacement between adjacent stories is the maximum story drift.

The maximum drift is observed for the 2nd floor

$$\Delta_1 = \delta_2 - \delta_1 = \mathbf{2.27in}$$

Correct answer is Option B.

31. Problem

Correct Answer: (A) 2 in

Explanation:

Per ASCE 7-16 page 109, table 12.12-1, allowable story drift for three story steel frame, risk category III

$$\Delta_a = 0.020 \cdot h_{sx}$$

$$\text{Story height, } h_{sx} = 11\text{ft} = 132 \text{ in}$$

$$\Delta_a = 0.020 \cdot 132 \text{ in}$$

$$\Delta_a = 2.64 \text{ in}$$

From section 12.12.1.1, page 109, for seismic design category

$$\text{Allowable drift} = \Delta_a / p$$

From section 12.3.4.2, page 98, $p = 1.3$ for extreme torsional irregularity

$$\text{Allowable drift} = 2.64 \text{ in} / 1.3 = 2.03 \text{ in} \approx \mathbf{2 \text{ in}}$$

Correct answer is option A.

32. Problem

Correct Answer: (C)

Explanation:

For pure torsion, the stress element only has shear stresses and no normal stresses.

For Mohr's circle, Principal stresses $\sigma_1 = \tau$, $\sigma_2 = -\tau$, and τ is the radius of the circle.

Hence, the correct answer is option C.

33. Problem

Correct Answer: (D) 625 kip*ft

Explanation:

Span of the beam, $L = 20\text{ft}$

Eccentricity, $e = 15\text{in}$

Load, $w = 25\text{ kips/ft}$

Total load $= w * L$

$$= 25\text{ kips/ft} * 20\text{ft} = 500\text{kips}$$

For beam restrained against torsion at one end only,

Torsional moment $= P * e$

$$= 500\text{ kips} * 15\text{ in} * (1/12\text{ ft/in})$$

$$= \mathbf{625\text{ kip*ft}}$$

Correct answer is option D.

34. Problem

Correct Answer: (B) 419.5 kip*ft

Explanation:

The reactions and internal forces in the beam due to a settlement of support A is equivalent to removing support A and applying an equivalent point load that causes a deflection equal to the settlement at point A.

Calculate the load 'P'

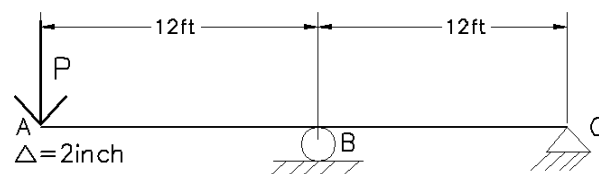
Ref to NCEES PE handbook page 247, Case 26

$$L = 12\text{ ft} = 144\text{ in}$$

$$a = 144\text{ in}$$

$$\Delta = 2\text{ in}$$

$$\Delta = P a^2 * (l+a) / (3EI)$$



$$P = \Delta * 3 EI / \{a^2 * (l+a)\}$$

$$P = 2 \text{ in} * 3 * 29000 \text{ ksi} * 1200 \text{ in}^4 / \{144 \text{ in} * 144 \text{ in} * (144 \text{ in} + 144 \text{ in})\}$$

$$P = 34.96 \text{ kips}$$

$$\text{Moment at support B} = P * 12 \text{ ft}$$

$$= 34.96 \text{ kip} * 12 \text{ ft}$$

$$= \mathbf{419.5 \text{ kip} * \text{ft}}$$

Correct answer is Option B

35. Problem

Correct Answer: (C) Option III

Explanation:

Ref to AISC 360-16, page 16.1-495 Table C-N5.4-1 (part 16 of AISC steel manual)

Groove welds require inspections for joint preparation prior to welding. Fillet welds require little to no joint preparation. Option III is the only figure that shows groove welds. Hence, Option C is the correct answer.

36. Problem

Correct Answer: (B)

Explanation:

Ref to IBC 2018 page 419, section 1705.5.2

Site built wooden trusses with heights of 60 inches or greater and trusses with spans 60 feet or greater are subjected to the requirements of special inspections. Per IBC 1704.2.5.1, fabricators approved to work without special inspection requirements are exempt from this. Hence the correct answer is Option B, site built metal plate connected King truss with a span 65 feet.

37. Problem

Correct Answer: (C)

Explanation:

Structural calculations for construction loads is not the responsibility of the general contractor. It is done by the structural engineer/ design professional responsible for the project. Hence Option C is not a part of the contractor's submittal package and is the correct answer.

38. Problem

Correct Answer: (D) 6 kips

Explanation:

Weight of beam, $W = 200 \text{ lbs/ft} \times 20 \text{ ft}$

$W = 4 \text{ kips}$

This acts at the center of the beam

Distance of point of action of beam weight from wire rope attachment point

$= 13 \text{ ft} + 6 \text{ ft} - 20 \text{ ft} / 2$

$= 9 \text{ ft}$

Let V be the vertical reaction at the column

Taking sum of moment at wire rope support point $= 0$

$W \times 9 \text{ ft} - V \times 6 \text{ ft} = 0$

$V = W \times 9 \text{ ft} / 6 \text{ ft}$

$V = 4 \text{ kips} \times 1.5$

$V = 6 \text{ kips}$

Correct answer is option D.

39. Problem

Correct Answer: (B) 12 in

Explanation:

Maximum inelastic response displacement of building 1

$$\delta_{M1} = 7 \text{ in}$$

Maximum inelastic response displacement of building 2

$$\delta_{M2} = 9 \text{ in}$$

Ref to ASCE 7-16, page 110, equation 12.12-2

Required structural separation

$$\delta_{MT} = \sqrt{\delta_{M1}^2 + \delta_{M2}^2}$$

$$\delta_{MT} = \sqrt{(7\text{in})^2 + (9\text{in})^2}$$

$$\delta_{MT} = 11.4 \text{ in} \approx \mathbf{12 \text{ in}}$$

Correct answer is option B.

40. Problem

Correct Answer: (D) 100 kips

Explanation:

Per OSHA, CFR Title 29, Part 1910, Subpart F, 1910.66 Appendix D, (b) (20) (iii)

The minimum factor of safety against the manufacturer's rated load shall be 10.

Required rated load

$$= \text{Static Load} * 10$$

$$= 10 \text{ kips} * 10$$

$$= \mathbf{100 \text{ kips}}$$
 Correct answer is Option D.

41. Problem

Correct Answer: (A) 2920 psf

Explanation:

Ref to NCEES PE handbook page 2 for density of water,

$$\gamma_w = 62.4 \text{ pcf}$$

Effective stress at 45 ft = Total stress - Pore water pressure

$$= \gamma_1 * h_1 + \gamma_2 * h_2 + \gamma_3 * h_3 - \gamma_w * h_w$$

$$= 120 \text{ pcf} * (75 \text{ ft} - 60 \text{ ft}) + 130 \text{ pcf} * (60 \text{ ft} - 55 \text{ ft}) + 128 \text{ pcf} * (55 \text{ ft} - 45 \text{ ft})$$

$$- 62.4 \text{ pcf} * (58 \text{ ft} - 45 \text{ ft})$$

$$= 2918.8 \text{ psf}$$

$$\approx \mathbf{2920 \text{ psf}}$$

Correct answer is option A.

42. Problem

Correct Answer: (B)

Explanation:

Sandy soils are more susceptible to liquefaction. It is caused due to decrease of effective stresses due to increase in pore water pressure. As the flowability of soil increases, the lateral earth pressure exerted also increases. Options A, C and D are all true.

Clayey soils, although less susceptible, are not immune to liquefaction. Option B is false and the answer.

43. Problem

Correct Answer: (B) 4.33 ksf

Explanation:

Ref to NCEES PE handbook page 90 section 3.4.2.1 for bearing capacity of square footings.

Depth of water table (25ft) is beyond (1.5*breadth of footing + depth)

Hence water table correction factor is not needed.

$$B_f = 6\text{ft}$$

$$L_f = 6\text{ ft (Square footing)}$$

From table in page 91, bearing capacity factors for $\Phi = 20$ degrees

$$N_c = 14.8$$

$$N_q = 6.4$$

$$N_\gamma = 5.4$$

From table in page 90, shape correction factors

$$S_c = 1 + (B_f/L_f) * (N_q/N_c)$$

$$S_c = 1 + 1 * (6.4/14.8)$$

$$S_c = 1.432$$

$$S_\gamma = 1 - 0.4 * (B_f/L_f)$$

$$S_\gamma = 1 - 0.4 * 1$$

$$S_\gamma = 0.6$$

$$S_q = 1 + (B_f/L_f) * \tan\Phi$$

$$S_q = 1 + 1 * \tan 20$$

$$S_q = 1.364$$

From equation in page 90, section 3.4.2.1

$$Q_{ult} = c * N_c * S_c + q * N_q * S_q + 0.5 * \gamma * B_f * N_\gamma * S_\gamma$$

$$Q_{ult} = 350 \text{ psf} * 14.8 * 1.432 + (115 \text{ pcf} * 5 \text{ ft}) * 6.4 * 1.364 + 0.5 * 115 \text{ pcf} * 6 \text{ ft} * 5.4 * 0.6$$

$$Q_{ult} = 13555.08 \text{ psf}$$

Calculate Q_{net}

$$Q_{net} = Q_{ult} - \gamma * D_f$$

$$Q_{net} = 13555.08 \text{ psf} - 115 \text{ pcf} * 5 \text{ ft} = 12.98 \text{ ksf}$$

Calculate allowable bearing pressure

$$Q_{all} = Q_{net} / \text{Factor of safety}$$

$$= 12.98 \text{ ksf} / 3 = \mathbf{4.33 \text{ ksf}}$$

Correct answer is option B.

44. Problem

Correct Answer: (C) Target air content 6 % and maximum w/c ratio 0.45

Explanation:

Ref to ACI 318-14, page 318 Table 19.3.1.1,

For frequent exposure to water, the exposure class is F2.

From page 323, table 19.3.2.1,

Maximum water cement ratio = 0.45 for class F2

From page 324, table 19.3.3.1, target air content = 6 % for class F2 and 3/4 in aggregates.

Hence the correct answer is Option C “Target air content 6 % and maximum w/c ratio 0.45”

45. Problem

Correct Answer: (A) 5.1

Explanation:

Ref to ACI 318-14 page 327, section 20.2.2.2.

Modulus of elasticity of non prestressed bars and wires

$E_s = 29000 \text{ ksi}$

Ref to ACI 318-14 page 315, section 19.2.2.1,

$E_c = w^{1.5} * 33 * \sqrt{f'_c}$

$E_c = 155^{1.5} * 33 * \sqrt{8000} \text{ psi}$

$E_c = (155^{1.5} * 33 * \sqrt{8000}) / 1000 \text{ ksi}$

$E_c = 5695.8 \text{ ksi}$

Modular ratio = E_s / E_c

$n = 29000 / 5695.8 = 5.1$

Correct answer is option A.

46. Problem

Correct Answer: (B) 4.46 ksi

Explanation:

Ref to PCI Handbook 7th edition, page 5-84, section 5.7.3 for loss due to elastic shortening

Ref to ACI 318-14 page 315, section 19.2.2.1, for modulus of elasticity of concrete

$$E_c = 57000 * \sqrt{f'_c}$$

$$E_{ci} = 57000 * \sqrt{5000}$$

$$E_{ci} = 4030.5 \text{ ksi}$$

From PCI handbook page 5-84, equation 5-65

For concentric strands, $e = 0$

$$f_{cir} = K_{cir} * (P_i / A_g)$$

$$f_{cir} = 0.9 * (350 \text{ kips} / 500 \text{ in}^2) = 0.63 \text{ ksi}$$

$E_{ps} = 28500 \text{ ksi}$ from page 5-84

From equation 5-64

$$E_s = K_{es} * E_{ps} * f_{cir} / E_{ci}$$

$$E_s = 1 * 28500 \text{ ksi} * 0.63 \text{ ksi} / 4030.5 \text{ ksi} = \mathbf{4.46 \text{ ksi}}$$

Correct answer is option B.

47. Problem

Correct Answer: (C) 9750 kip*in

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-67 section F13 for reduction in moment strength due to holes

$$F_u = 65 \text{ ksi}$$

$$F_y = 50 \text{ ksi}$$

Gross area of flange,

$$A_{fg} = 7 \text{ in} * 0.8 \text{ in} = 5.6 \text{ in}^2$$

Net area of flange for bolt holes on both sides of the web,

$$A_{fn} = 5.6 \text{ in}^2 - 2 * 0.875 \text{ in} * 0.8 \text{ in}$$

$$A_{fn} = 4.2 \text{ in}^2$$

$$F_y / F_u = 50 / 65 = 0.77 < 0.8.$$

$$\text{Hence } Y_t = 1$$

$$F_u * A_{fn} = 65 \text{ ksi} * 4.2 \text{ in}^2$$

$$F_u * A_{fn} = 273 \text{ kip}$$

$$Y_t * F_y * A_{fg} = 1 * 50 \text{ ksi} * 5.6 \text{ in}^2$$

$$Y_t * F_y * A_{fg} = 280 \text{ kip}$$

$F_u * A_{fn} < Y_t * F_y * A_{fg}$, Hence limit state of tensile rupture applies

From equation F13-1

$$M_n = F_u * (A_{fn} / A_{fg}) * S_x$$

$$= 65 \text{ ksi} * (4.2 \text{ in}^2 / 5.6 \text{ in}^2) * 200 \text{ in}^3$$

$$= \mathbf{9750 \text{ kip}\cdot\text{in}}$$

Correct answer is Option C.

48. Problem

Correct Answer: (D)

Explanation:

All options are false except Option D. Wood with higher density has a higher percentage of cell wall material which contributes more to overall shrinkage and swelling. Hence Option D is correct and the answer.

49. Problem

Correct Answer: (C) **600 psi**

Explanation:

Ref to NDS 2018 Supplement page 33 Table 4A

For Aspen No. 1, $F_c = 600 \text{ psi}$

(All adjustment values below are for F_c)

From page 32, Size factor for 10 in width and No 1, $C_f = 1$

From page 32, Wet service factor, $C_m = 0.8$

Also, check $F_c * C_f = 600 \text{ psi} * 1 < 750 \text{ psi}$

Thus, $C_m = 1$

From NDS 2018 page 11, Table 2.3.2, For live load, $C_d = 1$

From NDS 2018 page 11, Table 2.3.3, For 90°F, $C_t = 1$

From NDS 2018 page 29, Table 4.3.1,

$$F'_c = F_c * C_d * C_m * C_t * C_f * C_i * C_p$$

$$F'_c = 600 \text{ psi} * 1 * 1 * 1 * 1 * 1 * 1 = \mathbf{600 \text{ psi}} \quad \text{Correct answer is option C}$$

50. Problem

Correct Answer: (C) 43in

Explanation:

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Diameter of No. 6 bar, $d_b = 0.75$ in

Ref to TMS 402/602-16, page C-65 section 6.1.5.1.1 for development length

$K =$ smallest between cover, clear bar spacing between splices and $6*d_b$

$K = 3$ in

$\gamma = 1.3$ for No. 6 bar

$f_y = 60$ ksi

From equation 6-1

$$L_d = (0.13 * d_b^2 * f_y * \gamma) / (K * \sqrt{f'_m})$$

$$L_d = [0.13 * (0.75 \text{ in})^2 * 60000 * 1.3] / (3 \text{ in} * \sqrt{2000})$$

$$L_d = 42.5 \text{ in}$$

Correct answer is Option C, **43in**.

51. Problem

Correct Answer: (C) 17 ksi

Explanation:

Ref to TMS 402/602-16 page C-32, Table 4.2.2

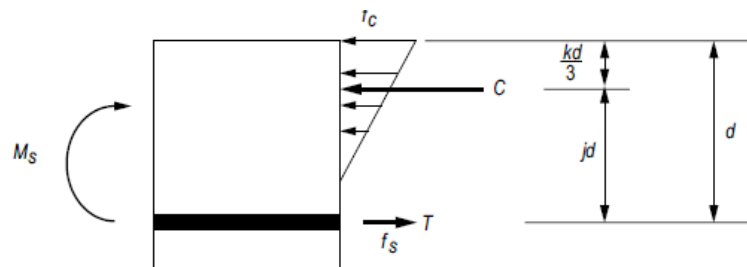
$$E_m = 900 * f'_m = 900 * 2000 \text{ psi} = 1800000 \text{ psi}$$

$$E_s = 29000000 \text{ psi}$$

$$\text{Modular ratio, } n = E_s / E_m = 29000000 \text{ psi} / 1800000 \text{ psi} = 16.11$$

Ref to NCEES PE handbook page 260, section 4.3.1

$$\text{Area of \#9 bar, } A_s = 1 \text{ in}^2$$



$$d = 24 \text{ in} - 4 \text{ in} = 20 \text{ in}$$

$$b_w = 7.625 \text{ in}$$

$$p = A_s / (b_w * d)$$

$$p = 1 \text{ in}^2 / (7.625 \text{ in} * 20 \text{ in}) = 0.00656$$

Calculate depth of N.A. from equation in NCEES PE handbook page 265, section 4.3.2.3

$$k = \sqrt{[(p*n)^2 + 2p*n]} - p*n$$

$$k = \sqrt{[(0.00656*16.11)^2 + 2*0.00656*16.11]} - 0.00656*16.11$$

$$k = 0.366$$

$$j = 1 - k/3$$

$$j = 1 - 0.366/3 = 0.878$$

$$f_s = M / (A_s * j * d)$$

$$f_s = 25 \text{ kip*ft} * (12 \text{ in/ft}) / (1 \text{ in}^2 * 0.878 * 20 \text{ in})$$

$$f_s = 17.08 \text{ ksi}$$

Hence, the correct answer is Option C, **17 ksi**.

52. Problem

Correct Answer: (A) Compact

Explanation:

Given $F_y = 50 \text{ ksi}$

Modulus of elasticity for steel, $E = 29000 \text{ ksi}$

Ref to AISC Steel manual, Part AISC 360-16, page 16.1-18, Table B4.1b Case 11,

For compact sections, compact limit

$$\lambda_p = 0.38 * \sqrt{E/F_y}$$

$$= 0.38 * \sqrt{(29000 \text{ ksi}/50 \text{ ksi})}$$

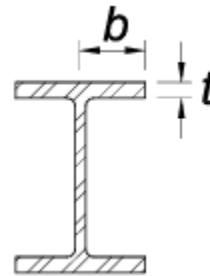
$$= 9.15$$

$$b/t$$

$$= (8 \text{ in} / 2) / 0.5 \text{ in}$$

$$= 8 < \lambda_p$$

Hence the flange is **compact** and the correct answer is Option A.



53. Problem

Correct Answer: (D) 5.5 in

Explanation:

Reactions due to uniformly distributed factored load on simply supported end is

$$R_u = w \cdot L / 2$$

$$R_u = 10 \text{ kip/ft} \cdot 25 \text{ ft} / 2 = 125 \text{ kip}$$

$$\Phi R_n = R_u = 125 \text{ kip}$$

From AISC steel manual 15th ed. page 9-48, Table 9-4,

$$\Phi R_1 = 74 \text{ kips}$$

$$\Phi R_2 = 23.5 \text{ kip/in}$$

$$\Phi R_3 = 90.9 \text{ kips}$$

$$\Phi R_4 = 6.25 \text{ kip/in}$$

From AISC Steel manual page 9-23, eq. 9-46a

For web local yielding, since load is throughout the span,

$$\Phi R_n = \Phi R_1 + l_b \cdot \Phi R_2$$

Rearranging,

$$l_b = (\Phi R_n - \Phi R_1) / \Phi R_2$$

$$l_b = (125 \text{ kips} - 74 \text{ kips}) / 23.5 \text{ kip/in}$$

$$l_b = 2.17 \text{ in}$$

From AISC Steel manual page 9-24, eq. 9-48a

For web local crippling, assuming $l_b/d \leq 0.2$

$$\Phi R_n = \Phi R_3 + l_b \cdot \Phi R_4$$

Rearranging,

$$l_b = (\Phi R_n - \Phi R_3) / \Phi R_4$$

$$l_b = (125 \text{ kips} - 90.9 \text{ kips}) / 6.25 \text{ kip/in}$$

$$l_b = 5.45 \text{ in}$$

Hence the second case governs.

Correct answer is Option D, **5.5 in**.

54. Problem

Correct Answer: (A) 48 kips

Explanation:

$$F_y = 50 \text{ ksi}$$

$$l_b = 1 \text{ in}$$

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-142, equation J10-1

For Flange local bending,

$$R_n = 6.25 t_f^2 F_y$$

$$R_n = 6.25 (0.51 \text{ in})^2 * 50 \text{ ksi}$$

$$R_n = 81.2 \text{ kips}$$

$$R_n / \Omega = 81.2 \text{ kips} / 1.67 = 48.6 \text{ kips}$$

From page 16.1-143, equation J10-2, for loads sufficiently away from supports,

For Web local yielding,

$$R_n = F_y t_w (5k + l_b)$$

$$R_n = 50 \text{ ksi} * 0.3 \text{ in} * (5 * 0.81 \text{ in} + 1 \text{ in})$$

$$R_n = 75.8 \text{ kips}$$

$$R_n / \Omega = 75.8 \text{ kips} / 1.5 = 50.5 \text{ kips}$$

Hence the 1st case governs allowable strength and the closest answer is Option A.

55. Problem

Correct Answer: (B) 10.5 ft

Explanation:

From TMS 402/602-16, page C-52, section 5.2.1.1.1,

Span of beams not built integrally with support shall be taken as clear span plus depth of beam and need not exceed the c/c distance between supports

Height of beam, $h = 2.5 \text{ ft}$

Clear span, $l = 8 \text{ ft}$

Effective span $= l + h$

$$= 8 \text{ ft} + 2.5 \text{ ft}$$

$$= \mathbf{10.5 \text{ ft}}$$

Correct answer is Option B.

56. Problem

Correct Answer: (C) 9

Explanation:

Unbraced length, $l_u = 6\text{ ft}$

From NDS-2018 supplement page 14, table 1B,

For nominal size 4X8, standar dressed size = 3.5X7.25

$$l_u / d = 6 * 12 \text{ in} / 7.25 \text{ in} = 9.9$$

From NDS-2018, page 16, Table 3.3.3, for uniformly distributed load

When $l_u / d > 7$

Effective length, $l_e = 1.63 * l_u + 3 * d$

$$l_e = 1.63 * 6 * 12 \text{ in} + 3 * 7.25 \text{ in}$$

$$l_e = 139.11 \text{ in}$$

From NDS 2018 page 17, eq 3.3-5

$$R_b = \sqrt{(l_e * d / b^2)}$$

$$R_b = \sqrt{[139.11 \text{ in} * 7.25 \text{ in} / (3.5 \text{ in})^2]}$$

$$R_b = 9.07$$

Correct answer is Option C, **9**.

57. Problem

Correct Answer: (C) 101 in

Explanation:

From ACI-318-14, page 68, Section 6.3.2.1 and Table 6.3.2.1,

Effective overhang is the least of

- a) $8 * \text{slab thickness} = 8 * 6 \text{ in} = 48 \text{ in}$
- b) $\text{Clear distance to adjacent web} / 2 = (9\text{ft} - 20\text{in}) / 2 = 44 \text{ in}$
- c) $\text{Clear span of beam} / 12 = 27 \text{ ft} / 8 = 40.5 \text{ in}$

Effective flange width for the center beam = beam width + 2*effective overhang

$$= 20 \text{ in} + 2 * 40.5 \text{ in} = \mathbf{101 \text{ in}}$$

Correct answer is option C.

58. Problem

Correct Answer: (B) 528 in²

Explanation:

From AISC steel manual page 15th ed. Page 16.1-140 and 16.1-141, section J8

The bearing frustum A2 is the maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area.

Loaded Plate area = 18 in X 16 in = 288 in² = A1

Pedestal area = 24 in X 24 in

Maximum dimension of pedestal = 24 in

Maximum dimension of plate = 18 in

24 in - 18 in = 6 in

Hence the area of bearing frustum geometrically similar to the plate

= (18 in + 6 in) X (16 in + 6 in)

= **528 in²**

Check $A2 < 4 A1$ $528 \text{ in}^2 < 4 * 288 \text{ in}^2$ O.K.

Hence correct answer is Option B, **528 in²**.

59. Problem

Correct Answer: (C) 1

Explanation:

All beams and columns seem to have hinges at the connection points with each other.

The center column has a hinge at the base and the frame can be assumed to be pin connected.

Also, X- bracing prevents any sidesway. Hence the effective length factor 'K' = 1.

Correct answer is Option C.

60. Problem

Correct Answer: (A) 0

Explanation:

First, assume that there is no tension in the wall and calculate the moment of inertia of the uncracked section without considering rebar.

Wall is fully grouted so the entire cross section is engaged.

L = 5 ft = 60 in

t = 7.625in

$$H = 8 \text{ ft} = 96 \text{ in}$$

$$\text{Area, } A = L \cdot t$$

$$A = 60 \text{ in} \cdot 7.625 \text{ in}$$

$$A = 457.5 \text{ in}^2$$

Section modulus

$$S = t \cdot L^2 / 6$$

$$S = 7.625 \text{ in} \cdot (60 \text{ in})^2 / 6 = 4575 \text{ in}^3$$

Total bending stress due to V

$$F_b = V \cdot H / S$$

$$F_b = 5 \text{ kips} \cdot 96 \text{ in} / 4575 \text{ in}^3 = 0.105 \text{ ksi}$$

Total compressive stress due to P

$$F_c = P \cdot L / A$$

$$F_c = (12 \text{ kip/ft} \cdot 5 \text{ ft}) / 457.5 \text{ in}^2$$

$$F_c = 0.131 \text{ ksi}$$

Minimum stress acting on the section

$$= F_c - F_b = 0.131 \text{ ksi} - 0.105 \text{ ksi} = 0.026 \text{ ksi (compressive, i.e. initial assumption valid)}$$

Hence there is no tension in the section and the correct answer is Option A, **0**.

61. Problem

Correct Answer: (B) 1322 kips

Explanation:

$$\text{Gross area, } A_g = 16 \text{ in} \times 16 \text{ in} = 256 \text{ in}^2$$

From ACI 318-14, page 507, Appendix A

$$\text{Area of \#8 bar} = 0.79 \text{ in}^2$$

$$A_{st} = 8 \cdot 0.79 \text{ in}^2 = 6.32 \text{ in}^2$$

From ACI 318-14, page 350, equation 22.4.2.2,

$$P_o = 0.85 \cdot f'_c \cdot (A_g - A_{st}) + f_y \cdot A_{st}$$

$$P_o = 0.85 \cdot 6 \text{ ksi} \cdot (256 \text{ in}^2 - 6.32 \text{ in}^2) + 60 \text{ ksi} \cdot 6.32 \text{ in}^2$$

$$P_o = 1652.6 \text{ kips}$$

From ACI 318-14, page 350, Table 22.4.2.1,

$$\text{For tied columns, } P_{n,\max} = 0.8 \cdot P_o$$

$$P_{n,\max} = 0.8 \cdot 1652.6 \text{ kips}$$

$$P_{n,\max} \approx \mathbf{1322 \text{ kips}}$$
 Correct answer is Option B.

62. Problem

Correct Answer: (D) 10.5 in²

Explanation:

Gross area, $A_g = 16 \text{ in} \times 16 \text{ in} = 256 \text{ in}^2$

Factored axial load, $P_u = 798 \text{ kips}$

Nominal axial strength, $P_n = P_u / \Phi$

$P_n = 798 \text{ kips} / 0.65 = 1227.7 \text{ kips}$

$K_n = P_n / (f'_c * A_g)$

$K_n = 1227.7 \text{ kips} / (4 \text{ ksi} * 256 \text{ in}^2)$

$K_n = 1227.7 \text{ kips} / (4 \text{ ksi} * 256 \text{ in}^2)$

$K_n = 1.199$

Factored moment, $M_u = 66.5 \text{ kip}\cdot\text{ft}$

Nominal moment, $M_n = M_u / \Phi$

$M_n = 66.5 \text{ kip}\cdot\text{ft} / 0.65 = 102.308 \text{ kip}\cdot\text{ft}$

$M_n = 1227.69 \text{ kip}\cdot\text{in}$

$R_n = P_n * e / (f'_c * A_g * h)$

$R_n = M_n / (f'_c * A_g * h)$

$R_n = 1227.69 \text{ kip}\cdot\text{in} / (4 \text{ ksi} * 256 \text{ in}^2 * 16 \text{ in})$

$R_n = 0.0749$

Plot the values of R_n and K_n in the interaction chart.

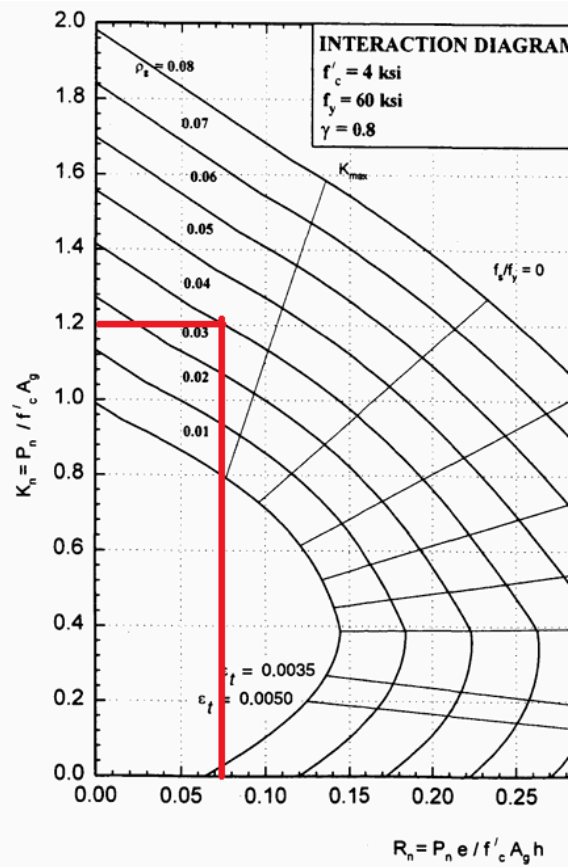
We get, ratio of Area of steel, $p = 0.04$

Required area of steel = $p * A_g$

$= 0.04 * 256 \text{ in}^2$

$= 10.24 \text{ in}^2$

Closest answer is Option D, **10.5 in²**.



63. Problem

Correct Answer: (B) 4ft

Explanation:

Ref to AISC steel manual 15th ed., part ASIC 360-16, page 16.1-94, section I3.1a

Effective width on either side of beam centerline is lesser of

- a) $1/8$ th of c/c between supports = $(1/8) * 16\text{ft} = 2\text{ft}$
- b) $1/2$ distance to centerline of adjacent beam = $(1/2) * 6\text{ft} = 3\text{ft}$
- c) Distance to slab edge = 3ft

Hence (a) controls and total effective width

$$= 2\text{ft} + 2\text{ft} = \mathbf{4\text{ft}}$$

Correct answer is option B.

64. Problem

Correct Answer: (B) 446 kip*ft

Explanation:

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length, number of studs, $n = 64/2 = 32$

$$\Sigma Qn = 16.28 \text{ kip} * 32 = 520.96 \text{ kips}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-7,

$$a = \Sigma Qn / (0.85 * f'_c * b) = 520.96 \text{ kips} / [0.85 * 4.25 \text{ ksi} * 6\text{ft} * (12\text{in}/\text{ft})] = 2 \text{ in}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-6,

$y_2 = Y_{\text{con}} - a/2$, where Y_{con} is the distance between top of steel beam to top of concrete

$$y_2 = (3\text{in} + 2.5 \text{ in}) - 2\text{in}/2 = 4.5 \text{ in}$$

From AISC Steel manual 15th edition, page 3-174 and 3-175, Table 3-19,

For W18X50, $y_2 = 4.5 \text{ in}$ and $\Sigma Qn = 521 \text{ kips}$, $M_{\text{allowable}} = \mathbf{446 \text{ kip*ft}}$

Correct answer is Option B.

65. Problem

Correct Answer: (B) 5 kips

Explanation:

Span of the truss, $L = 20$ ft

Depth of the truss, $d = 5$ ft

Factored Load, $P = 5$ kips

Max factored moment acting on the span with point load at the center

$$= P * L / 4$$

$$= 5 \text{ kips} * 20 \text{ ft} / 4$$

$$= 25 \text{ kip*ft}$$

Max design tension at the bottom chord

$$T = P / d$$

$$T = 25 \text{ kip*ft} / 5 \text{ ft}$$

$$T = \mathbf{5 \text{ kips}}$$

Correct answer is Option B.

66. Problem

Correct Answer: (C) 1000 kip*ft

Explanation:

For strong -column-weak-beam approach,

Per ACI 318-14, page 281, Equation 18.7.3.2

$$\Sigma M_{nc} \geq (6/5) \Sigma M_{nb}$$

$$\text{i.e. } \Sigma M_{nb} \leq (5/6) \Sigma M_{nc}$$

$$\Sigma M_{nb} \leq (5/6) * 1200 \text{ kip*ft}$$

$$\Sigma M_{nb} \leq \mathbf{1000 \text{ kip*ft}}$$

Hence the correct answer is Option C.

67. Problem

Correct Answer: (A) 13.2 kips

Explanation:

Ref to NCEES PE handbook page 257, section 4.2.1 for shear in eccentrically loaded bolts

No. of bolts, $n = 4$

Due to symmetry location of centroid of bolt group from bottom left bolt ,

$$x = 3\text{ in } y = 2\text{ in}$$

Eccentric moment acting on the group, $M = P * (18\text{ in} + 6\text{ in}/2) = 21 * P$

Moment of inertia of bolt group (Area of all the bolts is assumed equal)

$$I_x = \sum x^2 = 4 * 3\text{ in} * 3\text{ in} = 36\text{ in}^2 \quad I_y = \sum y^2 = 4 * 2\text{ in} * 2\text{ in} = 16\text{ in}^2$$

$$I_p = I_x + I_y = 36 + 16 = 52\text{ in}^2$$

Vertical load on each bolt due to direct shear = $P / 4 = 0.25 P$

Vertical load on each bolt due to moment = $(M/I_p) * x = 21 P / 52 * 3 = 1.212 P$

Total vertical shear = $0.25 P + 1.212 P = 1.462 P$

Horizontal shear on each bolt due to moment = $(M/I_p) * y = 21 * P / 52 * 2 = 0.8077 P$

$$\text{Total shear on each bolt} = \sqrt{(1.462 P)^2 + (0.8077 P)^2} = 1.67 P$$

Allowable shear in bolts = 22 kips

$$1.67 * P = 22\text{ kips}$$

$P = 22\text{ kips} / 1.67 = \mathbf{13.2\text{ kips}}$ Correct answer is Option A.

68. Problem

Correct Answer: (B) 7.87 kips

Explanation:

Thickness of filler plate, $t = 3/8\text{ in}$

From AISC steel manual 15th ed, page 16.1-139 , Section J5.2 (a)

Modification factor for shear strength of bolts

$$= 1 - 0.4 * (t - 0.25)$$

$$= 1 - 0.4 * (3/8 - 0.25)$$

$$= 0.95 > 0.85\text{ O.K.}$$

From AISC steel manual 15th ed page 7-22, Table 7-1

For 5/8 inch diameter A307 bolts in double shear,

Allowable shear = 8.29 kips

For the connection with filler, allowable bolt shear with modification factor
 $= 0.95 * 8.29 \text{ kips} = \mathbf{7.87 \text{ kips}}$

Correct answer is Option B.

69. Problem

Correct Answer: (C) 359 kips

Explanation:

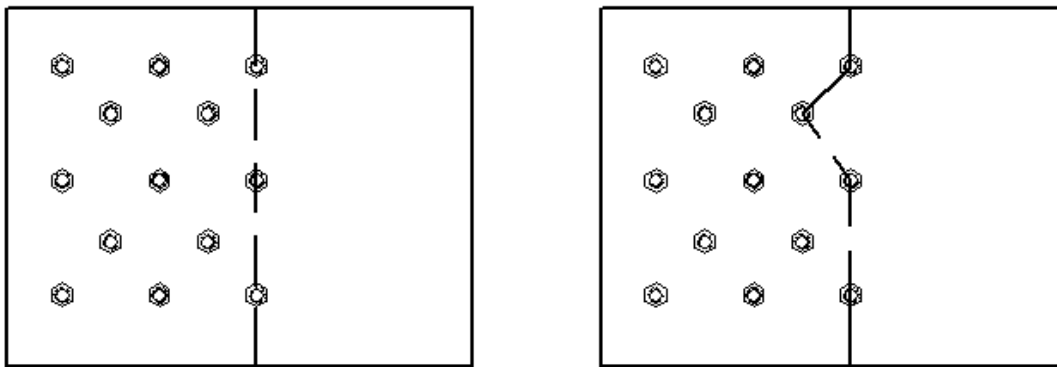
For net area, per AISC steel manual 15th ed. page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal hole dimension ,

$$= 13/16 \text{ in} + 1/16 \text{ in} = 0.875 \text{ in}$$

Longitudinal center to center spacing, $s = 3 \text{ in}$

Transverse spacing, $g = 2.5 \text{ in}$

The plates can fail in tensile rupture in two ways as shown below



Per AISC steel manual 15th ed. page 16.1-20, section B4.3b

$$\text{Additional length for each diagonal stagger} = s^2 / 4g = 3 \text{ in}^2 / (4 * 2.5 \text{ in}) = 0.9 \text{ in}$$

$$\text{For two diagonal staggers, additional length due to staggered path} = 0.9 \text{ in} * 2 = 1.8 \text{ in}$$

$$\text{Reduction in length of staggered path due to additional hole} = 0.875 \text{ in} < 1.8 \text{ in}$$

Hence staggered path with 4 holes does not govern, straight path with 3 holes governs

For straight path,

$$\text{Net area, } A_n = (15 \text{ in} - 3 * 0.875 \text{ in}) * 0.5 \text{ in}$$

$$A_n = 6.188 \text{ in}^2$$

$$\text{Nominal tensile rupture strength} = A_n * F_u$$

$$= 6.188 \text{ in}^2 * 58 \text{ ksi}$$

$$\approx \mathbf{359 \text{ kips}}$$

Correct answer is Option C.

70. Problem

Correct Answer: (A) 130.5 kips

Explanation:

Diameter of pin, $d = 2.875$ in

Hole diameter, $d_h = 3$ in

Thickness of plate, $t = 1$ in

From AISC steel manual 15th ed, page 16.1-29 and 16.1-31, Section D5.1

$$b_e = 2 \cdot t + 0.63 = 2 \cdot 1 \text{ in} + 0.63 = 2.63 \text{ in}$$

' b_e ' must be less than $3 \text{ in} - d_h/2 = 3 \text{ in} - 1.5 \text{ in} = 1.5 \text{ in}$, governs

$$b_e = 1.5 \text{ in}$$

$$a = 4.5 \text{ in} - d_h/2 = 4.5 \text{ in} - 3 \text{ in}/2 = 3 \text{ in}$$

$$A_s = 2 \cdot t \cdot (a + d/2)$$

$$A_s = 2 \cdot 1 \text{ in} \cdot (3 \text{ in} + 2.875 \text{ in}/2) = 8.875 \text{ in}^2$$

For tensile rupture in net area,

$$\Phi \cdot F_u \cdot (2 \cdot t \cdot b_e) = 0.75 \cdot 58 \text{ ksi} \cdot (2 \cdot 1 \text{ in} \cdot 1.5 \text{ in}) = \mathbf{130.5 \text{ kips}} \text{ (governs)}$$

For shear rupture in net area,

$$\Phi \cdot 0.6 \cdot F_u \cdot A_s = 0.75 \cdot 0.6 \cdot 58 \text{ ksi} \cdot 8.875 \text{ in}^2 = 231.6 \text{ kips}$$

From AISC steel manual 15th ed, page 16.1-28, Equation D2-1

$$\text{Strength in gross tension yielding} = \Phi \cdot F_y \cdot A_g = 0.9 \cdot 36 \text{ ksi} \cdot 6 \text{ in} \cdot 1 \text{ in} = 194.4 \text{ kips}$$

Correct answer is Option A.

71. Problem

Correct Answer: (D) 70.9 kips

Explanation:

Length of welds (single line), $L = 6\text{ in}$

Total vertical load in connection = P

Total moment in the connection = $P \cdot 3\text{ in}$

From AISC steel manual 15th ed. Page 8-14, Fig 8-6,

For single line horizontal weld, $I = L \cdot (dy)^2$

For two lines, $I = 2 \cdot L \cdot (dy)^2 = 2 \cdot 6\text{ in} \cdot (6\text{ in}/2)^2 = 108\text{ in}^3$

Section modulus, $S = I / (L/2) = 108\text{ in}^3 / 3\text{ in} = 36\text{ in}^2$

Shear per unit length of weld, $V = P / (2 \cdot L) = P/12 = 0.0833 P$

Max tension in the weld, $T = \text{Moment} / S = P \cdot 3 / 36 = P/12 = 0.0833 P$

Resultant per unit length = $\sqrt{(V^2 + T^2)} = \sqrt{[(0.0833 P)^2 + (0.0833 P)^2]} = 0.1178 P$

Here, both T and V are acting transverse to the weld axis.

From AISC steel manual 15th ed. Page 8-9, "Effect of Load angle" for transversely loaded welds, 50% increase can be taken

Size of 3/8 inch weld in 16th of an inch, $3/8\text{ in} = 6/16\text{ in}$. Hence $D = 6$

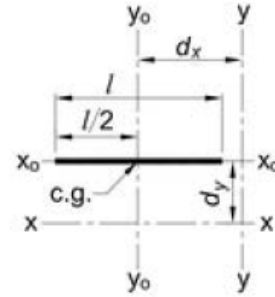
From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n/\Omega = 0.928 \cdot D \cdot l$

Resultant = Allowable stress per unit length of weld including load angle effect

Resultant = $0.928 \cdot D \cdot 1.5$

$0.1178 P = 0.928 \cdot 6 \cdot 1.5$

$P = 70.9\text{ kips}$ Correct answer is Option D.



72. Problem

Correct Answer: (D) 35.3 kips

Explanation:

From AISC steel manual 15th ed. Page 16.1-133 and 16.1-134, section J3.7,

When the required stress, in either shear or tension, is less than or equal to 30% of the corresponding available stress, the effects of combined stress need not be investigated.

From AISC steel manual 15th ed. Page 7-22, Table 7-1

Allowable shear strength of 1 inch diameter Group A bolts in single shear and threads excluded from shear plane,

$V_r = 26.7\text{ kips}$

Acting shear, $V = 5$ kips

$$V/V_r = 5 \text{ kips} / 26.7 \text{ kips} = 18.7 \% < 30\%$$

Hence no reduction in tension strength required.

From AISC steel manual 15th ed. Page 7-23, Table 7-2

Allowable tensile strength of Group A bolts, 1 inch diameter = **35.3 kips**

Correct answer is Option D.

73. Problem

Correct Answer: (B)

Explanation:

All options are true about the lack of fusion in welds except Option B.

Lack of fusion significantly affects the fatigue life during stress reversals but also significantly affects the static strength. Hence Option B is partially incorrect and is the answer.

74. Problem

Correct Answer: (D) **45 in**

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #6 bar, $d_b = 0.75$ in

From ACI 318-14, page 417, Table 25.4.2.4

$\Psi_t = 1.3$ for top bars with more than 12 in. of concrete below.

From ACI 318-14, page 415, Table 25.4.2.2, development length for #6 bar

$$L_d = (f_y * \Psi_t * \Psi_e) / (25 * \lambda * \sqrt{f'_c}) * d_b$$

$$L_d = (60000 * 1.3 * 1) / (25 * 1 * \sqrt{4000}) * 0.75 \text{ in} = 37 \text{ in}$$

From ACI 318-14, page 433, Section 25.6.1.5

Development length for individual bars within a bundle, in tension or compression, shall be that of the individual bar, increased 20 percent for a three-bar bundle.

$$L_d = 1.2 * 37 \text{ in} = 44.4 \text{ in}$$

Correct answer is Option D, **45 in.**

75. Problem

Correct Answer: (B)

Explanation:

Footings need to be placed below the frost line to prevent damage from freezing and thawing cycles.

Insulating the soil against frost effects is not the correct measure. The changes in footing size won't help against cycles of freezing and thawing. Pile foundation is a very expensive counter measure and is not the most appropriate solution.

Hence correct answer is Option B.

76. Problem

Correct Answer: (C) 0.55 in

Explanation:

Lateral load = 200 kips

Total moment arm

$$= 2\text{ft} + 1.5\text{ft} + 7\text{ ft} = 10.5\text{ ft}$$

The moment will be resisted by a couple acting at the center of the two footings.

Center to center distance between the footings,

$$= 12\text{ft} - 1\text{ft}/2 - 1\text{ft}/2 = 11\text{ ft}$$

Maximum thrust acting in the footing

$$= 200\text{ kips} \times 10.5\text{ ft} / 11\text{ ft}$$

$$= 190.9\text{ kips}$$

$$\text{Maximum bearing pressure} = 190.9\text{ kips} / (4\text{ft} \times 4\text{ft})$$

$$= 190.9\text{ kips} / (48\text{ in} \times 48\text{ in}) = 0.0829\text{ kip/in}^2 = 82.9\text{ lb/in}^2$$

$$\text{Modulus of subgrade reaction, } k = 150\text{ lb/in}^3$$

$$k = \text{Bearing pressure} / \text{settlement}$$

$$\text{Settlement} = \text{Bearing pressure} / k = [82.9\text{ lbs/in}^2] / [150\text{ lb/in}^3] = \mathbf{0.55\text{ in}}$$

Correct answer is option C

77. Problem

Correct Answer: (B) 716 kips

Explanation:

The column is located at the edge of the footing.

Supporting surface is not wider on all sides than the loaded area and the bearing frustum cannot be developed. Hence, only the area of column will be used for bearing calculation.

From ACI 318-14, page 379, Table 22.8.3.2 case (c)

$$B_n = 0.85 f'_c A_1$$

$$B_n = 0.85 * 4 \text{ ksi} * (18 \text{ in} * 18 \text{ in})$$

$$B_n = 1101.6 \text{ kips}$$

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.65$ for bearing

Design bearing capacity

$$= \Phi * B_n$$

$$= 0.65 * 1101.6 \text{ kips} \approx \mathbf{716 \text{ kips}}$$

Correct answer is Option B.

78. Problem

Correct Answer: (C)

Explanation:

Option A is false because end bearing piles are not always shorter than friction piles. The length depends on geotechnical conditions.

The bearing capacity of skin friction piles are affected by the pile material so option B is false.

Both end bearing and friction piles can be used in cohesive or cohesionless soil so Option D is also false.

Negative skin friction affects both end bearing and friction piles. Hence Option C is true and is the answer.

79. Problem

Correct Answer: (D) 448 kips

Explanation:

$$\text{Area of rebar, } A_s = 2 * 4 \text{ in}^2 = 8 \text{ in}^2$$

$$\text{Area of concrete, } A_c = (\pi * D^2 / 4) - A_s$$

$$A_c = (\pi * 12 \text{ in} * 12 \text{ in} / 4) - 8 \text{ in}^2$$

$$A_c = 105.1 \text{ in}^2$$

From IBC 2018, page 447, Table 1810.3.2.6

Allowable stress in concrete cast in rock = $0.33 f'_c$

$$f_{ca} = 0.33 * 6 \text{ ksi} = 1.98 \text{ ksi}$$

Allowable stress in compression reinforcement = $0.4 * f_y$

$$f_{sa} = 0.4 * 75 \text{ ksi} = 30 \text{ ksi}$$

$$\text{Allowable load in pile} = f_{ca} * A_c + f_{sa} * A_s$$

$$= 1.98 \text{ ksi} * 105.1 \text{ in}^2 + 30 \text{ ksi} * 8 \text{ in}^2$$

$$= 448.1 \text{ kips}$$

Hence the correct answer is Option D, **448 kips**.

80. Problem

Correct Answer: (A) 25 kip*ft

Explanation:

Maximum intensity of earth pressure at the base of the stem

$$= K_a * \gamma * H = 0.33 * 120 \text{ pcf} * 10 \text{ ft} = 396 \text{ psf}$$

$$\text{Max load acting on the bottom 1 ft strip} = 396 \text{ psf} * 1 \text{ ft} = 0.396 \text{ kip/ft}$$

From ACI 318-14, page 60, section 5.3.8,

load factor for earth pressure = 1.6,

$$\text{factored load} = 1.6 * 0.396 \text{ kip/ft} = 0.634 \text{ kip/ft}$$

The stem is considered a continuous beam system with a uniform loading, $w_u = 0.634 \text{ kip/ft}$

and clear span, $l_n = 20 \text{ ft}$

From ACI 318-14, page 70, Table 6.5.2, for interior segment of wall with more than two

spans, max negative moment = $w_u * l_n^2 / 10$

$$= (0.634 \text{ kip/ft}) * (20 \text{ ft})^2 / 10 = 25.36 \text{ kip*ft}$$

Correct answer is option A, **25 kip*ft**.